Learned selection for exact Feynman integral reduction with Kira

Rushil Saraswat

October 14, 2026

Abstract

We study learned choices of seeds, elimination rows and equations during exact Feynman integral reduction with Kira. We include prediction, algebra, checks and retries in the measured time. Original tests give an 8.6% saving from seed selection with supplied equations and a 5.7% saving from equation selection in four native graph structures. Row selection gives no reliable saving and is omitted. A later global cost predictor gives no reliable extra gain over native controls. We therefore fit seventy candidates with separate models for seven graph structures. The selected route gives 6.3% less time in 720 reserved trials, with a time ratio of 0.9368 to native choices and 0.9441 to a fixed strategy per graph. Every trial passes exact checks. Combining compatible requests gives a separate 48.3% saving through shared reduction. Exact recurrences and reuse give a 543.8-fold saving in a restricted vacuum family through twelve loops. The native studies cover known graph structures at one through three loops; the results retain their stated inputs, controls and timing conditions.

1 Introduction

Higher-order perturbative corrections to scattering amplitudes and particle decay amplitudes contain integrals over loop momenta. Higher loop orders produce many integrals with related numerators and denominators, and algebraic identities can express them in terms of a smaller set of master integrals [1, 2].

Kira performs this reduction by generating integration-by-parts (IBP) identities and solving the resulting linear equations [3, 4]. It returns the coefficient of each master integral in a requested target. Evaluating the masters is a separate calculation. Kira already uses methods that reduce unnecessary equations and algebraic work [5]. Our aim is to test whether learned models can further reduce that work.

We study three choices: which seeds generate equations, which row eliminates an integral, and which individual equations enter the calculation. We select model types by complete reduction time, including inference and exact checks. Earlier work has also used machine-assisted search to find seed-selection rules [6].

We first define the notation used in this paper and explain how the three choices enter Kira. We then describe the data, each model and its tests. We use these results to study complete-cost model selection and shared reductions for compatible requests. We then test separate policies for each graph with new seed strategies. We compare full calculation times because fewer equations or accurate predictions alone do not show that a reduction is faster. The appendices give the integral definitions, derivations and detailed measurement rules.

2 How Kira reduces Feynman integrals

An integral family fixes the propagators and external scalar products. Changing the integer powers gives different integrals in that family [1, 4]:

$$I_{\boldsymbol{a}}(\boldsymbol{x}, d) = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{i\pi^{d/2}} \prod_{i=1}^{N} D_i^{-a_i}, \qquad D_i = q_i^2 - m_i^2 + i0. \tag{1}$$

Here L is the number of loops, ℓ_h is loop momentum h, and N is the number of inverse propagators, including auxiliary entries. The momentum q_i is a linear combination of loop and external momenta, and m_i is a particle mass. The symbol $i0$ gives the Feynman prescription. We use dimensional regularization with $d = 4 - 2\epsilon$, where ϵ is the regulator. The vector $\boldsymbol{x}$ contains the independent momentum invariants and squared masses.

The integer a_i specifies the power of D_i: a positive value gives a denominator, zero removes the factor, and a negative value gives a numerator. Auxiliary inverse propagators provide scalar products that the physical propagators do not cover. A *sector* groups integrals with the same set of positive indices [4].

To obtain an IBP equation, choose an index vector $\boldsymbol{\sigma}$, called a *seed*, and a loop or external momentum v. Differentiate with respect to loop momentum ℓ_k and integrate. In dimensional regularization, the total derivative gives [2]

$$0 = \int \prod_{h} \frac{d^d \ell_h}{i\pi^{d/2}} \frac{\partial}{\partial \ell_k^\mu} \left(v^\mu \prod_i D_i^{-\sigma_i} \right) = \sum_j A_{\boldsymbol{\sigma},o;j}(\boldsymbol{x}, d) I_{\boldsymbol{a}^{(j)}}. \tag{2}$$

The operator label o specifies k and v, and j labels the integrals produced by the derivative. Each $A_{\boldsymbol{\sigma},o;j}$ is an algebraic coefficient. Appendix A expands the derivative and shows how an equation eliminates an integral. Several operators can act on one seed, so one seed generally produces several equations.

Kira generates equations within specified index limits, uses symmetries and removes redundant equations. It then eliminates integrals in order and substitutes the resulting relations back into the targets [4, 5]. Kira can use exact arithmetic modulo a prime to select equations or reconstruct rational functions. For a prime p, residues modulo p form a finite field. Kira can also reduce symbolic rational functions directly [4, 7]. A target can be one integral or a sum $T = \sum_u c_u I_{\boldsymbol{a}^{(u)}}$. Its reduction is

$$T = \sum_{m=1}^{n_M} r_{Tm}(\boldsymbol{x}, d) M_m, \qquad r_{Tm} \in \mathbb{Q}(\boldsymbol{x}, d). \tag{3}$$

Here c_u is a target coefficient, M_m is master integral m, and n_M is the number of masters. The notation $\mathbb{Q}(\boldsymbol{x}, d)$ means rational functions of the invariants, squared masses and dimension with rational-number coefficients. We check that every target reduces to the prescribed master basis, then compare its coefficients with an independently obtained exact reduction.

Each equation is one row of the coefficient matrix, and each integral is one column. Elimination removes an integral by combining rows. Native Kira uses its built-in selection rules. Table 1 places the three models in this procedure. Each model can save different work, and each prediction has a cost. Exact algebra still determines the coefficients. If the selected equations cannot reduce a target to the prescribed basis, the calculation retries with native Kira choices. We call this retry a *fallback*.

Table 1: The three decisions and the work each model can avoid.

Model	Decision	Possible saving
Seed selection	Which index vectors supply IBP identities?	Generate fewer groups of equations.
Row selection	Which allowed row eliminates the next integral?	Reduce intermediate expressions and operations.
Equation selection	Which seed–operator pairs are useful?	Avoid constructing unnecessary coefficients.

3 Training data and fair comparisons

To train the original models, we construct targets from checked master relations and exact source identities. Let $E_q = \sum_i A_{qi} I_i = 0$ be identity q. We form

$$T = \sum_m r_m M_m + \sum_q \lambda_q E_q, \qquad \lambda_q, r_m \in \mathbb{Q}(\boldsymbol{x}, d). \tag{4}$$

Expanding the right-hand side gives a target whose master coefficients are known. We save the seeds, operators and construction steps. Appendix B proves the relation. The target can be a sum of integrals; it is a single indexed integral only when the expansion has that form. The saved path gives one sufficient reduction without proving minimum cost.

Table 2 gives the original collection size. It contains 57,979 master entries counted separately in each family. It covers loop orders one through twelve, but its higher-loop additions belong to the restricted vacuum family in Appendix D. Many family instances differ only in their parameters. These counts therefore do not describe 17,100 different scattering graphs. The test set contains 148,494 single integrals and 541,888 integral expressions. Appendix F defines the families, and Appendix G explains the master-basis checks.

Related mass profiles stay together in either training or testing at all loop orders. The seed and row networks each receive twenty further training passes over the expanded training set. They retain their earlier fitted parameters and optimizer state. Some older test profiles had appeared in earlier evaluations, so this test concerns reserved parameter groups rather than entirely new graphs. Later native tests use separate training, validation and test cases. Validation chooses the settings; test data do not change them.

Each time comparison uses the same targets, master basis and resources for all methods. Full time includes preparation, prediction, algebra, checks and every fallback. Offline training and reference discovery remain outside this time. The main comparisons use five repetitions and compare median times. Most native tests use the same modified Kira 3.1 executable for learned and native choices. The cumulative test uses the saved original executable as its baseline. Appendix E defines the averages and uncertainty intervals.

Table 2: Original constructed data. Family instances include parameter variations of the same graph.

Part	Family instances	Targets
Training	11,963	1,610,890
Testing	5,137	690,382
Total	17,100	2,301,272

4 Model 1: seed selection

The original seed network receives the target terms, family, master indices and a finite candidate seed list. It converts each term into 32 learned numerical features. It then combines these features, ordered pairs of terms, family information and master information to score each seed. The shared term representation and attention operation follow Refs. [8, 9]. Appendix C gives the architecture. The network has 28,097 fitted parameters and assigns each seed a score:

$$p_b = \frac{1}{1 + e^{-g_\theta(T,F,\mathcal{M},\boldsymbol{\sigma}_b)}}, \qquad \widehat{\mathcal{S}}(T) = \{\boldsymbol{\sigma}_b \in \mathcal{C} : p_b \geq \tfrac{1}{2}\}. \tag{5}$$

Here F is the family, $\mathcal{M}$ is the master list, $\mathcal{C}$ is the candidate list, and g_θ is the network output with fitted parameters θ. The score estimates whether a seed belongs to the saved sufficient set. Several seeds can be selected, so their scores do not need to sum to one. Training uses binary cross-entropy and AdamW [10]; Appendix C derives the loss. Prediction receives neither the final master coefficients nor later reduction steps.

On the 690,382 test targets, precision is 95.750%, recall is 96.892%, and F_1 is 96.318%. Complete seed sets match the saved sets for 84.332% of targets. Kira completes 652,517 reductions with the predicted sets, giving 94.515% sufficiency; it completes all targets with the full pool. These measures answer different questions because a different seed set can still reduce a target. Appendix E defines the metrics. We report precision and recall because the many zero labels can make overall accuracy misleading [11].

The separate time test uses existing identity pools from twelve reserved one-loop box instances. It uses unchanged weights and threshold 0.01, fixed before the comparison. The earlier label and sufficiency tests use threshold 1/2. Both the learned selector and a distance-based rule receive the same basis check and full-pool fallback. For single targets, the learned/full-pool time ratio is 0.914, with a descriptive 95% interval of $[0.871, 0.958]$. This means 8.6% less time. The ratio to the distance rule is 0.913 with interval $[0.855, 0.972]$, or 8.7% less time. Every method completes all five repetitions for these twelve cases.

For sixteen-target batches, the ratio is 0.952 with interval $[0.772, 1.148]$, so the measurements do not establish a consistent saving. Across the planned seed comparison, 345 trials finish exactly, three are incomplete, and twelve repetitions are skipped. The timing summary uses complete paired cases. This test excludes the original generation of the identity pool. A separate native test that includes IBP generation gives 1.096 with interval $[0.963, 1.275]$. The original seed model therefore has a demonstrated saving only in the stated supplied-equation test. Section 7 studies new seed trees trained directly on scattering reductions.

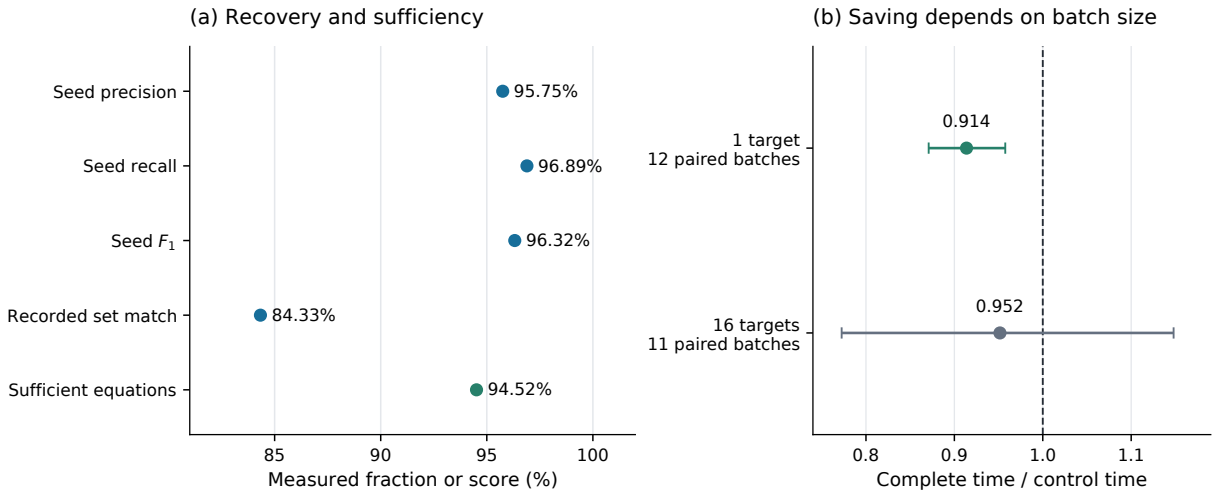


Figure 1: Seed prediction accuracy and reduction speed measure different outcomes. Left: results on the original reserved targets. Right: learned time divided by full-pool time. Values below one are faster. The single-target interval lies below one; the sixteen-target interval crosses it.

5 Model 2: row selection

The row network receives the current matrix and the permitted source rows for the next elimination. Its 132 inputs describe the pivot integral, nonzero row entries, family, matrix size and sampled cancellations. It gives each action 32 learned features, combines information across actions, and uses a 64-component hidden layer to produce a score. A softmax normalizes the exponentials of the scores into probabilities, and the model chooses the largest. The network has 10,529 parameters. Exact arithmetic performs the chosen operation. Appendix A derives that operation, and Appendix C defines the network.

The original training data follow one changing target row with fixed source identities. On 2,558,472 reserved states with a choice, the network selects the saved action in 95.845% of states. A local deterministic rule reaches 99.473%. Complete paths match the saved master row at three finite-field points for 690,302 of 690,382 targets. These sampled checks do not prove symbolic equality. Including prediction, the network takes 17.657 times the local rule's time.

We also use the fixed network inside Kira's changing symbolic matrix while preserving the allowed integral order. Across 84 exactly checked trials, full time is 1.305 times stock Kira time. Later, smaller policies learn from measured algebraic cost and use compiled prediction and shared features. Their latest row-only ratio is 0.982 with interval $[0.959, 1.003]$. Adding the seed policy gives 1.100 with interval $[0.966, 1.279]$ against native choices.

These intervals do not establish a reliable saving, as Figure 2 shows. We therefore omit learned row selection from the retained method and let Kira choose the rows. We preserve these negative results because agreement with a saved reduction path did not produce faster algebra.

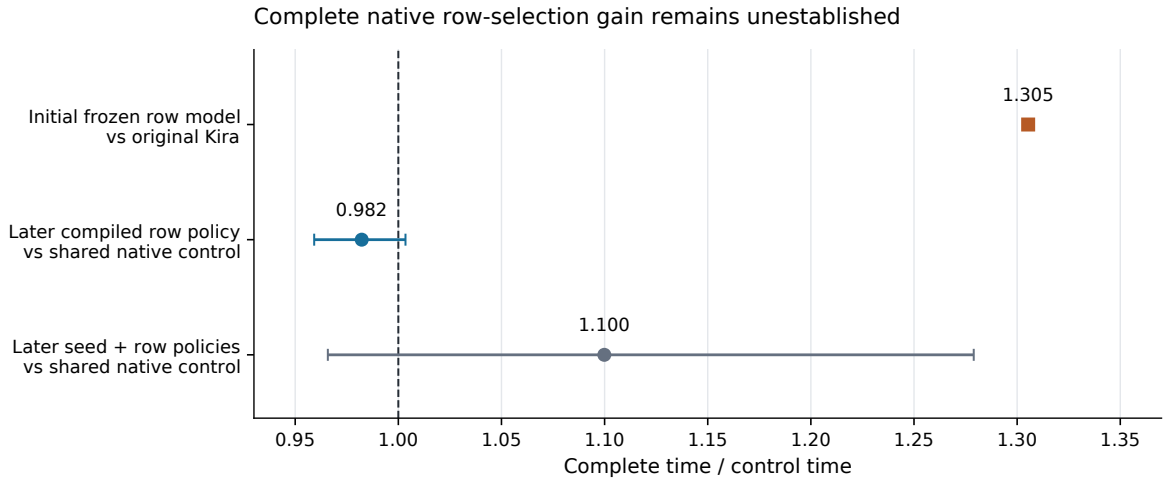


Figure 2: The row model gives no reliable full-time saving. The first model is slower, and the later row-only interval includes equal time. The original summary provides no interval for the first result. These comparisons use different model versions.

6 Model 3: equation selection

The equation model scores each seed–operator pair before Kira constructs its symbolic coefficients. It uses seventeen features that describe the family, targets, seed indices and operator. They include the loop count, target count, extra propagator powers, numerator rank and distance between seed and target indices. It needs no equation coefficients. A classification tree [12] estimates whether the equation belongs to the useful part of a recorded native reduction. The tree follows feature-threshold decisions to a leaf, which stores the score. Kira keeps its own treatment of symmetries and masters.

The original fit uses 153,124 candidate equations, including 12,270 positive labels, from fourteen batches in eight parameter profiles. The graphs are a two-loop sunrise, planar double box, three-loop vacuum and three-loop ladder. Validation chooses between trees of depth eight and twelve, with 295 and 1,335 nodes, at threshold 0.01. The double box uses the deeper tree; the vacuum and ladder use the shallower tree; the 21-target sunrise case retains the preceding equation policy. Validation requires lower full and preparation times with no fallback before activating a filter. Adding learned seed or row choices fails these criteria, so this test keeps native seed and row choices.

The test uses eight new parameter profiles and fourteen target batches. Masses and momentum invariants have rational values while d remains symbolic, so the coefficients lie in $\mathbb{Q}(d)$. Five repetitions of four methods give 280 full calculations. Native choices, the early equation model and the distance-based control use the same executable improvements. A fourth method uses the preceding release. This comparison separates the model's choices from improvements shared by all methods.

The equation model reduces full time by 5.7% and equation-preparation time by 36.2%. Their ratios are 0.943 with interval $[0.915, 0.970]$ and 0.638 with interval $[0.546, 0.734]$. Preparation includes modular generation, selection, modular solving and symbolic equation regeneration. Other stages limit the full-time saving. All 280 calculations finish on their first attempt. Independent exact checks cover 94,970 coefficient comparisons and eight applications given only the prescribed basis.

We next apply the fixed depth-twelve tree to a massless nonplanar two-loop box, massless pentagon and box with one external virtuality. These structures were absent from its training data. Appendix F gives their propagators and invariants. The nonplanar family follows the massless

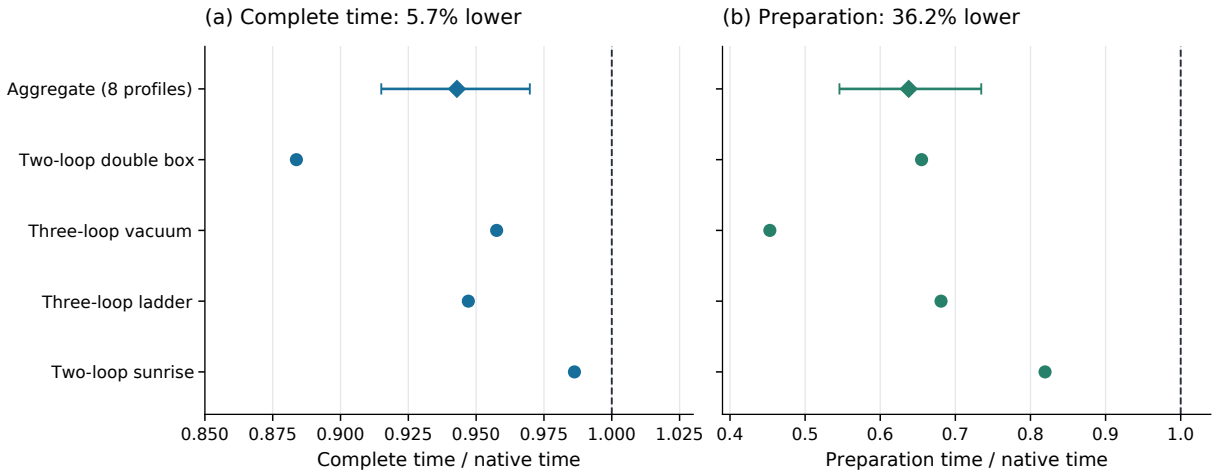


Figure 3: Equation selection saves more preparation time than full calculation time. Each ratio compares the model with native choices in the same executable. The panels use different labelled ranges. The aggregate intervals describe variation across the eight profiles.

limit in Ref. [4]. Four batches use rational kinematics and symbolic d. Three further batches keep all independent invariants symbolic, giving coefficient fields $\mathbb{Q}(d, s, t)$, $\mathbb{Q}(d, s_{12}, s_{23}, s_{34}, s_{45}, s_{51})$ and $\mathbb{Q}(d, s, t, v)$. Here s, t are four-point invariants, $s_{i,i+1}$ are adjacent five-point invariants, and v is an external squared momentum.

The seven batches contain 88 targets and give seventy comparison calculations. All results pass exact checks, but all 35 filtered calculations require native fallback. Their average time ratio is 1.252, which means 25.2% more time. Checks cover 5,600 coefficient pairs. Enlarged seed regions agree in all seven cases, and new rational substitutions agree for the three symbolic cases. This establishes correct completion through fallback. It also shows that the original filter does not save time on these inputs. Figure 4 preserves this result from before the new training.

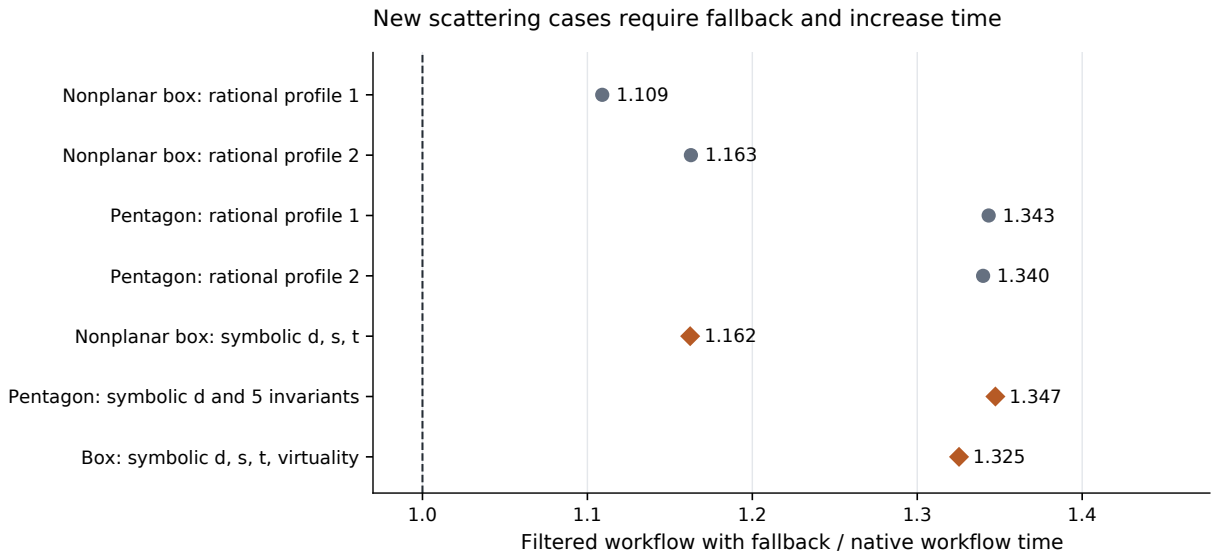


Figure 4: The original equation model is slower on the new scattering structures. Each point compares five-run median times, including fallback. All 35 filtered calculations require a retry. This historical test uses the original weights, before additional training.

7 Training on the missing scattering structures

The earlier equation model had no training examples from the nonplanar box, pentagon or box with external virtuality. To test whether training on these structures helps, we fit a new seed tree and equation tree for each structure. We preserve the original models and their tests. Appendix C defines the new labels and models.

The study uses separate training, validation and test batches. Table 3 gives their sizes. Rational batches use different invariant values in each split. Symbolic batches keep all independent invariants symbolic and use disjoint target lists within the symbolic split. The graph structures are shared across splits, so this tests adaptation within these structures.

Table 3: The additional scattering study. Each of the three structures contributes equally to the batch counts.

Part	Rational batches	Symbolic batches	Targets
Training	6	3	108
Validation	3	3	60
Testing	6	3	108

Each tree has maximum depth twelve. The seed trees use sixteen features and assign the same score to all operators of a seed. The equation trees use seventeen features, including the operator index. Kira enumerates its usual seed region, and the selected model filters equations before coefficient construction. Both methods preserve native elimination. These are new tree models fitted on native reductions; the original neural networks remain unchanged.

Validation compares three score thresholds and fixes the best setting before testing. The test compares native choices, the original equation tree, the new seed tree and the new equation tree on identical targets. It includes prediction, exact algebra, coefficient checks, cleanup and fallback. Reference discovery, training and later record archiving remain outside the timer. Five repetitions give 180 test calculations. Separate applications receive a master basis without access to target reference coefficients, and their returned coefficients are checked afterward.

The new fit uses 59,706 equation records and 4,674 grouped seed records. These counts include

repeated candidate patterns in different training batches. Table 4 gives the sizes of the fitted trees. No test record enters these fits.

Table 4: Additional training records and tree sizes. Each graph uses three training batches.

Structure	Seed records	Seed nodes	Equation records	Equation nodes
Nonplanar box	4,326	67	56,238	819
Pentagon	258	15	2,838	179
Box with virtuality	90	11	630	81

All 180 test calculations pass exact coefficient checks, covering 9,880 returned coefficient entries. All eighteen basis-only applications also pass their separate coefficient checks. Each new selector completes all 45 first attempts without fallback. The original equation model requires fallback in all 45 trials on these same new test cases.

Removing fallback improves the learned methods relative to the original filter, but does not establish a saving against native Kira. The new seed model's full-time ratio is 1.027, with descriptive 95% interval $[0.995, 1.086]$. The new equation model's ratio is 1.017, with interval $[0.988, 1.054]$. Their measured average times are therefore 2.7% and 1.7% higher than native time. The original filter's ratio on this new test is 1.368, or 36.8% more time. This differs from the earlier 25.2% result because the parameter points and symbolic targets differ.

Table 5: New reserved test results. Each value uses five repetitions. Ratios divide model time by native time; values below one are faster.

Case	Native (s)	Original tree	New seeds	New equations
Nonplanar box (13)	4.163	1.293	1.023	0.980
Nonplanar box (14)	4.140	1.221	1.002	0.974
Nonplanar box (symbolic)	4.458	1.233	0.983	1.010
Pentagon (13)	1.005	1.439	0.963	1.016
Pentagon (14)	0.965	1.436	1.069	1.038
Pentagon (symbolic)	1.120	1.356	0.958	0.979
Box with virtuality (13)	0.826	1.339	0.953	0.991
Box with virtuality (14)	0.763	1.529	1.152	1.124
Box with virtuality (symbolic)	0.729	1.499	1.166	1.052
Geometric mean ratio	—	1.368	1.027	1.017

Table 5 and Figures 5–6 show the distinction between enough equations and faster reduction. Some cases become faster and others become slower. We keep native choices as the application default for these scattering cases because the reserved test gives no reliable overall saving. The trained models remain available for further comparisons.

The result supports adding examples from the missing structures to improve equation sufficiency. It does not prove that missing data were the only cause of the original failure. This study also fits separate trees for each graph and changes the seed-model architecture. It tests adaptation within three structures, rather than transfer to arbitrary new graphs or higher loops. The timing intervals resample the three graph groups and describe this small comparison. They do not establish performance over all scattering integrals.

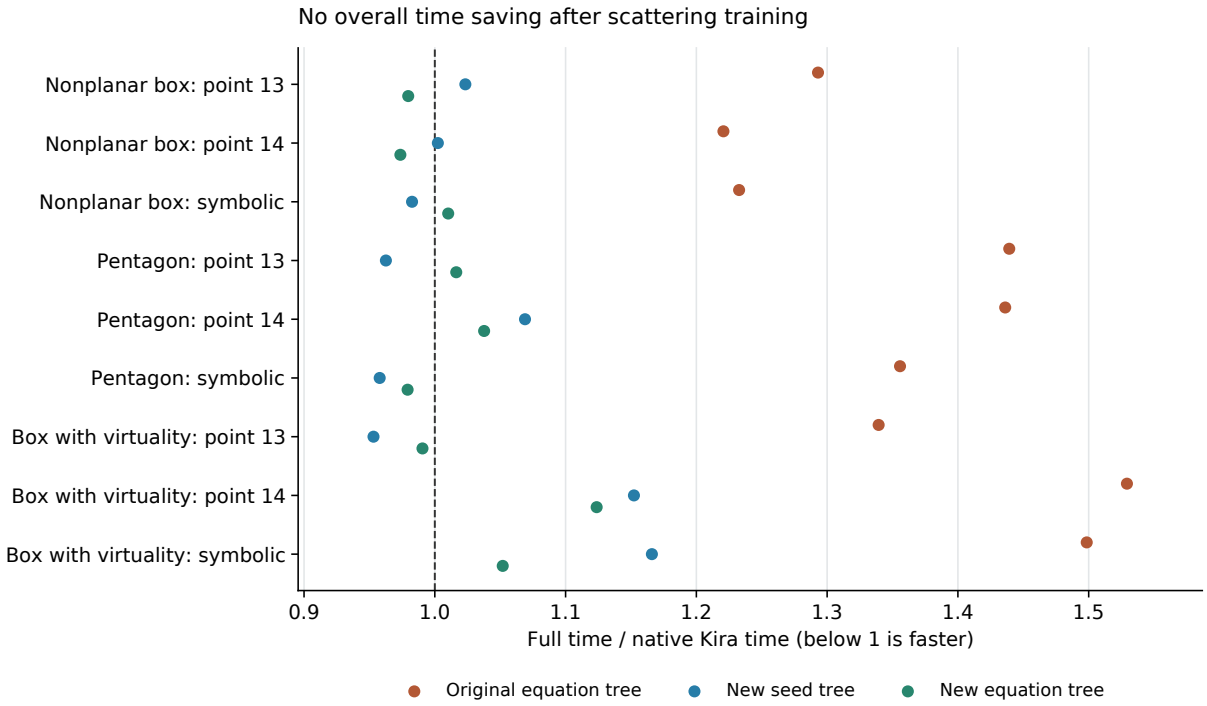


Figure 5: Training on the three structures removes much of the original filter's overhead, but gives no overall saving against native Kira. Points compare five-run median full times, including prediction, checks and fallback. The dashed line marks equal time. No per-case uncertainty interval is inferred from five repetitions.

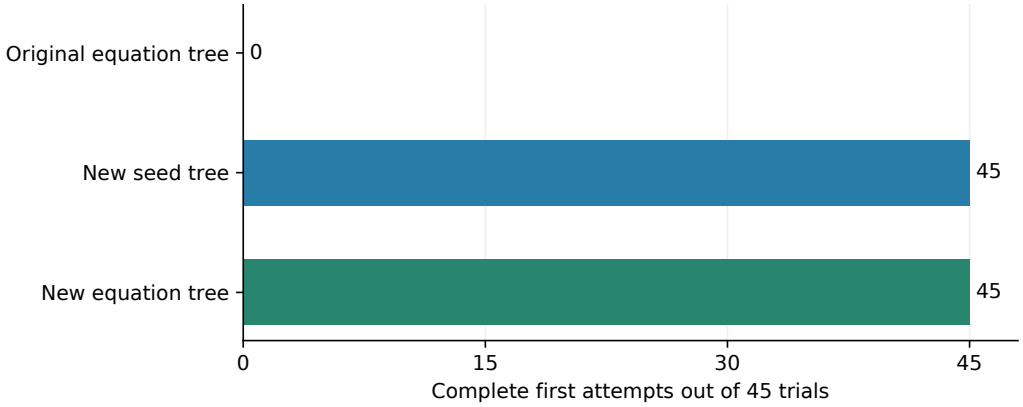


Figure 6: The new selectors provide sufficient equations on every first attempt. Each method has 45 trials across nine batches. Native fallback lets the original filter eventually complete, and its added time remains in the comparison above.

8 Choosing a complete reduction strategy

The preceding tests show that supplying enough equations does not by itself ensure a faster reduction. We therefore train a model on complete reduction times. For a target batch and a prescribed master basis, the model chooses the initial seed region, an optional seed or equation filter, and an integral ordering. Kira then performs exact reduction. The choice occurs once for the batch, which avoids repeated model calls during elimination.

We also combine the earlier equation filter with an existing improvement to temporary data handling. This changes execution cost while preserving the equations and arithmetic. A separate control uses this combined path with native decisions. Comparing the model with that control measures what the learned choice adds to the integration improvement.

Let $B = (r_{\max}, s_{\max}, q_{\max})$ contain the largest positive index sum, numerator rank and dot count among the targets. A dot is one denominator power above one. Appendix I gives their definitions and the cost-model derivations. Table 6 lists the candidate strategies. Kira provides the seed limits and integral-ordering options used here [13]. Ordering five changes the sector priority while keeping the same within-sector preference as ordering one. Every comparison uses the same prescribed master basis. For the four earlier multiloop graphs, the seed-filter option uses native choices because no graph-specific seed tree is available in this route. The equation option uses the earlier tree for those graphs and the new scattering trees for the three additional structures.

Table 6: Strategies available to the complete-cost model. Negative seed bounds are replaced by zero. Native row choices remain active.

Strategy	Initial seed region and selection
Integrated native	Bounds B; native choices
Smaller region	Bounds $B - (1, 1, 1)$; native choices
Fewer dots	Bounds $B - (1, 0, 1)$; native choices
Equation filter	Bounds B; saved equation tree
Seed filter	Bounds B; saved seed tree where available
Different ordering	Bounds B; Kira ordering five
Larger region	Bounds $B + (1, 1, 1)$; native choices

The larger region can avoid an unsuccessful first attempt when a target needs equations beyond its own index limits. The smaller regions can avoid unnecessary generation when fewer equations suffice. If a proposed region or filter fails the basis check, the calculation retries with native choices at B, then at $B + (1, 1, 1)$ when needed. The native control also includes enlargement after an unsuccessful first attempt. Failed attempts remain in the time used for both training and testing.

The training label for each strategy is the logarithm of its complete-time ratio to integrated native choices. The model receives eighteen features that describe the graph, symbolic variables, target count, index powers and sectors. We compare a constant predictor, six regression trees, three ridge-regression models and two nearest-neighbour models. Regression trees follow the squared-error construction in Ref. [12]. Appendix I specifies each model. Validation selects ridge regression with penalty $\lambda = 10$ and margin $\gamma = 0$ before the held-out tests. Its validation time ratio to integrated native choices is 0.993. The best fixed strategy from training is integrated native choices. Thus the integrated and best-fixed controls repeat the same configuration in separate trials, and differences between their times reflect execution variation. The seed and equation trees retain their earlier weights, and the original row model remains omitted.

The study has 31 training batches, 17 validation batches and 31 test batches. It covers the sunrise, planar box, three-loop vacuum, three-loop ladder, nonplanar box, pentagon and box with external virtuality. Rational batches use separate invariant or mass values in each split.

Additional symbolic batches use different target lists. These are new inputs within known graph structures. The native tests cover one through three loops; the separate exact recurrence results through twelve loops retain their restricted vacuum scope.

Each training and validation cost uses three repetitions. Testing compares the previous native path, integrated native choices, the best fixed strategy from training, and the learned strategy with five repetitions. A running controller starts a fresh Kira process for every attempt. The timer includes input preparation, prediction, native IBP generation, algebra, exact coefficient checks, cleanup and every retry. Offline training, enlarged reference reductions and later evidence archiving are outside the timer. Separate applications receive only the target and master basis; their coefficients are checked afterward. The learned application uses native choices for unmatched graph structures or target complexity outside the training range. This guard also applies when requests are combined. Cold command startup is recorded in the application checks; the comparative timings describe the running controller.

8.1 Full reduction results

All 620 timed workflows pass exact checks of 59,640 coefficient entries. All 31 applications supplied with a master basis also pass their later coefficient checks. Each timed method makes fifteen fallback attempts. The new selector therefore changes some initial strategies without removing the remaining insufficiency failures.

The learned-to-previous-native time ratio is 0.9903, with descriptive interval $[0.9796, 0.9996]$, or about 1.0% less time. Its ratio to integrated native choices is 0.9965 with interval $[0.9875, 1.0070]$. Its ratio to the best-fixed control is 1.0026 with interval $[0.9979, 1.0081]$. Both latter intervals include equal time, so the test establishes no additional full-time saving from the learned choices. The integrated and best-fixed controls have ratios 0.9938 and 0.9878 to the preceding native path. Their difference, despite using the same configuration, also shows the scale of timing variation in this comparison.

Table 7 gives the results by graph. We average logarithmic ratios within each graph and then give each of the seven graphs equal weight. Appendix I derives this average and defines the descriptive intervals. The smallest measured ratio for the learned route is 0.9654, for the three-loop vacuum. The ladder and nonplanar box take slightly more time. These measurements do not establish a significant speedup across all the tested integral types.

Table 7: Complete-cost results. Native seconds are geometric means of batch median times. The other entries are geometric mean time ratios; a value below one means less time.

Graph	Native (s)	Integrated/native	Learned/native	Learned/integrated
Sunrise	0.502	1.0123	0.9932	0.9811
Planar box	1.621	0.9906	0.9771	0.9864
Three-loop vacuum	0.726	0.9745	0.9654	0.9907
Three-loop ladder	3.535	1.0146	1.0013	0.9869
Nonplanar box	3.119	1.0075	1.0073	0.9998
Pentagon	0.599	0.9758	0.9942	1.0189
Box with virtuality	0.484	0.9824	0.9944	1.0122

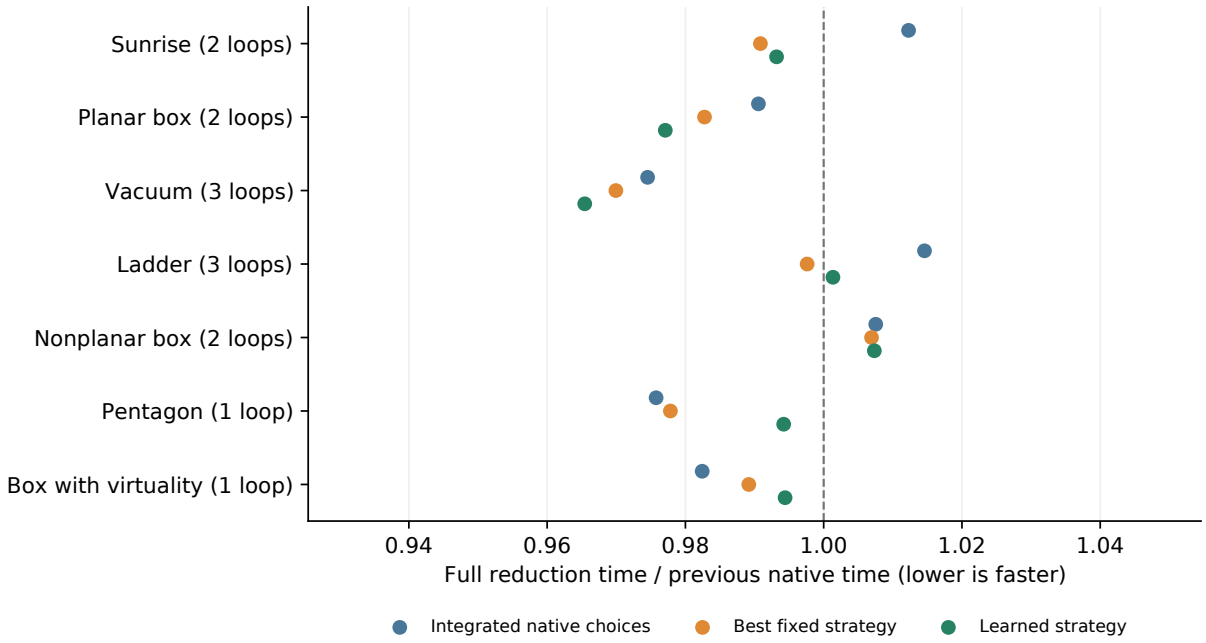


Figure 7: Learned choices and the native controls have similar full times. The best-fixed and integrated controls use the same configuration in separate trials. The plot shows variation across graph structures; the text gives the aggregate uncertainty intervals. The axis focuses on the measured range and retains the equal-time line.

8.2 Sharing reductions between compatible requests

The small differences between single-batch strategies motivate a change in how requests reach Kira. When two target lists have the same family and kinematics and are available together, we combine their targets, reduce the union once, and return the coefficient rows for each list. This shares the family preparation and algebra. Appendix I specifies the ten request groups and derives the output separation from linearity. The model and its settings remain fixed.

Five repetitions of four methods give 200 complete workflows, and all pass exact coefficient checks. Joint native reduction has time ratio 0.5252 to two separate native reductions. The combined integration with native choices gives ratio 0.5175 with interval $[0.5073, 0.5346]$, or 48.3% less time. This is a speedup of 1.93 with interval $[1.87, 1.97]$. Every tested graph group takes less time with joint reduction. The integrated native ratios range from 0.5040 to 0.5679 across the seven graphs, corresponding to savings of 43.2–49.6%.

The joint learned route gives ratio 0.5152 to separate native requests. Its ratio to joint integrated native choices is 0.9955, with interval $[0.9810, 1.0058]$. This interval includes equal time. Table 8 and Figure 8 therefore attribute the main measured gain to sharing the reduction. The model adds no reliable gain over the integrated native control. The standard application uses the best fixed strategy from training, which retains native choices. The learned strategy remains an optional route. Ten checks of each joint application supply only the basis, then verify the returned coefficients independently. Ten further checks exercise the single-batch command. These application checks add to the 31 basis-only applications in the main comparison.

Table 8: Joint-request time ratios and descriptive intervals over the seven graph groups. Each group contains two compatible target requests.

Comparison	Time ratio	95% interval
Joint native / separate native	0.5252	[0.5130, 0.5420]
Joint integrated / separate native	0.5175	[0.5073, 0.5346]
Joint learned / separate native	0.5152	[0.5022, 0.5322]
Joint learned / joint native	0.9810	[0.9657, 0.9946]
Joint learned / joint integrated	0.9955	[0.9810, 1.0058]

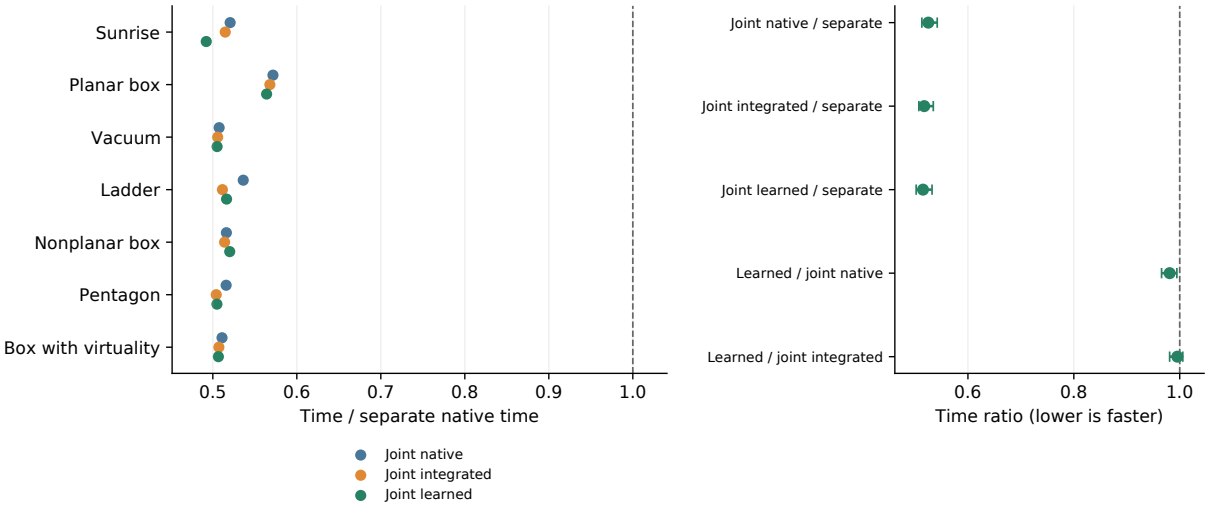


Figure 8: Combining compatible requests gives the main saving. Left: ratios by graph relative to two separate native reductions. Right: aggregate ratios and descriptive intervals. Comparing learned choices with joint integrated native choices gives no reliable extra saving. The panels retain the equal-time reference.

The joint comparison reuses reserved parameter points from the main test. It provides a new comparison on that population. Transfer to additional graph structures remains untested. The saving requires compatible requests that can share one calculation. These measurements neither establish a saving for an isolated arbitrary integral nor extend native IBP performance claims to general scattering graphs beyond three loops.

9 Choosing a policy for each integral type

The preceding study used one cost predictor across seven graph structures. We now train separate predictors and enlarge the set of possible seed strategies. This lets a target-dependent choice depend on the graph's own measured costs. The new candidates include restricting the maximal target sectors and reducing numerator seeding in lower sectors, using Kira's truncation option [5]. The comparison retains exact algebra, master-basis checks, and fallback. Every method first enforces a valid positive-index bound for its declared sectors.

We fit ten candidates for each graph, giving seventy fitted models. They include regression trees, a bagged tree ensemble, regularized linear models, and nearest-neighbour models. The model receives nine features of the target indices and family and predicts complete-time ratios for nine strategies. The response includes failed attempts and retries. We train on 36 batches, choose settings on 18 validation batches, and reserve 36 batches at new rational parameter values for testing. The dimension remains symbolic. The targets include denominators raised to power three, lower sectors, and numerator indices -2 on auxiliary entries. Appendix J gives the exact definitions, model fits, and selection rule. The final training and validation records contain 2,798.5 seconds of measured reductions, and fitting all seventy candidates takes 0.86 seconds. The label total excludes reference construction, development, and replaced measurements. These offline costs remain outside the reserved reduction times.

The route enables a new cost model only when validation shows at least five percent less time than the fastest available control. This threshold is fixed before testing. The controls are native choices, a deterministic sector-and-truncation rule, and one fixed strategy per graph selected from training. The last control can include the older equation filter. Validation activates a new cost model for 3 of the seven graphs and retains a control strategy for the others. Unknown graph structures and target features outside the training range use native choices. A chosen strategy can still fail its basis check; the route then retries, and the comparison includes that cost.

Five repetitions of four methods give 720 reserved timing trials. All trials pass exact coefficient checks. Giving each graph equal weight, the selected route has geometric mean full-time ratio 0.9368 with descriptive interval $[0.8867, 0.9878]$, corresponding to 6.3% less time. Its ratio to the fixed choice per graph is 0.9441 with descriptive interval $[0.8992, 0.9863]$. The aggregate interval lies below equal time. Table 9 gives each graph's result. The selected route has a time ratio below one for 5 of the seven graph groups. These are measurements within the tested input groups, not a speed guarantee for every integral in those graphs. Across its 180 timed trials, the selected route makes 0 retry attempts; native choices make 40 retries in the matched 180 trials. The time ratios include these retries.

The enabled trees give 14.2%, 12.4%, and 13.8% less time for the planar box, ladder, and nonplanar box, respectively. The vacuum and virtuality box show 3.7% and 0.8% more time. Both of those groups use a retained control throughout. The vacuum route selects native settings, so its measured difference includes routing overhead and timing variation.

Table 9: Results for each graph. Times are geometric means of the batch median times, in seconds. A ratio below one means less complete time. The route column names the enabled cost model or retained control.

Graph	Route	Native	Selected	Ratio
Sunrise	Stronger truncation	0.825	0.811	0.983
Planar box	Depth-two tree	3.410	2.925	0.858
Vacuum	Native	1.033	1.072	1.037
Ladder	Depth-one tree	5.144	4.506	0.876
Nonplanar box	Depth-two tree	6.234	5.376	0.862
Pentagon	Target sectors	1.040	0.989	0.951
Virtuality box	Sectors and truncation	0.808	0.815	1.008

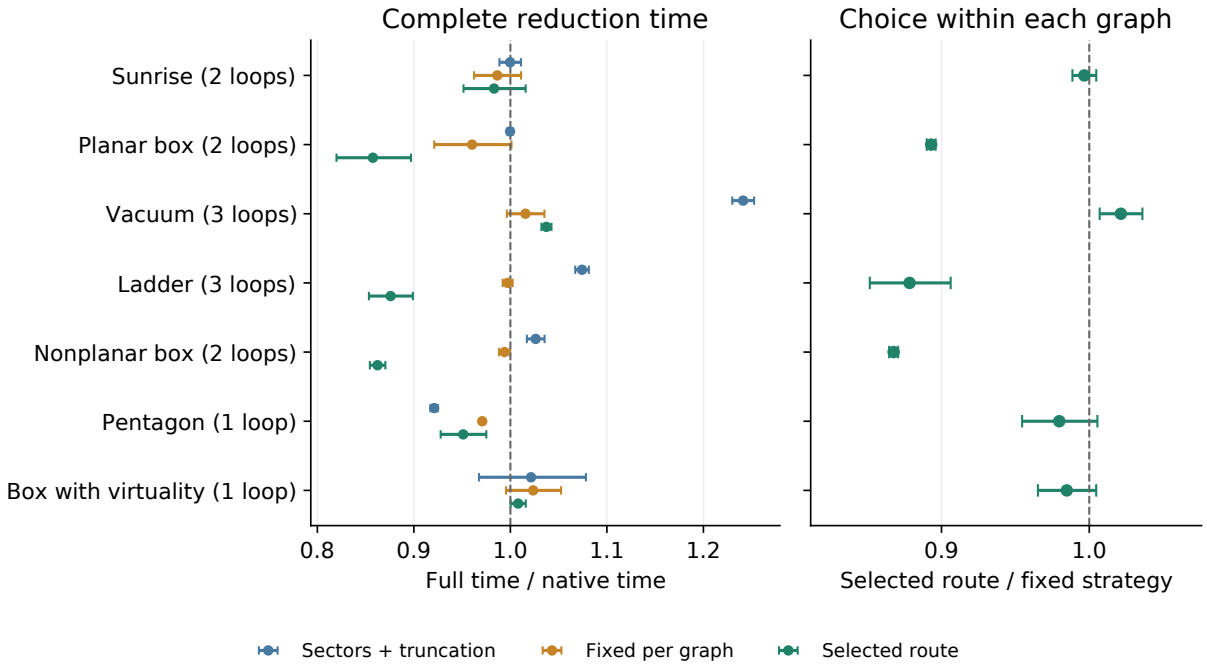


Figure 9: The effect of choosing a separate policy for each graph. Left: full-time ratios to native choices. Right: the selected route relative to one fixed strategy per graph. Bars show the range across two reserved parameter profiles; they are not significance intervals. Both panels retain the equal-time reference.

The timing trials check 14,040 coefficient entries. We also complete 36 applications supplied only with the master basis and seven further application checks, then compare their coefficients independently. These additional applications check 945 coefficients. The new study uses the same seven graph structures as before, at one through three loops. Its target patterns occur in training, and it tests new kinematic profiles. General scattering beyond three loops and new target structures remain outside this comparison. The per-graph models have two training parameter profiles; the fixed range guard and reserved checks limit the claims accordingly.

10 Overall results

The original comparisons support learned seeds for the tested supplied-equation requests and learned equations for the tested native families. They give no reliable saving from learned row selection. Table 10 separates these results because they use different inputs and time different amounts of work. Multiplying the speedups would not give a measured combined speedup. The original native equation experiment also gives no established gain from activating both selectors together. Section 7 reports the separate scattering-training study. Section 8 then tests complete-cost selection and joint requests. Its 48.3% saving comes mainly from sharing the calculation; the learned cost model adds no reliable saving over the integrated native control. This result is a direct comparison of identical paired requests. The separate graph-specific study in Section 9 gives 6.3% less time relative to native choices and ratio 0.9441 to the fixed strategy per graph. Its gains cannot be multiplied by the joint-request saving because the two studies use different requests.

Table 10: Summary of separate comparisons. Each row uses its stated inputs and baseline. The studies provide no measured product of these speedups.

Method	Tested calculation	Result for full time
Seed network	One target from an existing equation pool	8.6% less time
Equation tree	Native reduction in four trained graph structures	5.7% less time
Row policy	Native symbolic elimination	No reliable saving
Original equation tree	Three new scattering structures	25.2% more time
Complete-cost model	Seven known graph structures, loops 1–3	No reliable gain over native controls
Policies per graph	Higher powers and lower sectors, loops 1–3	6.3% less time
Joint integrated reduction	Two compatible requests in seven graph structures	48.3% less time
Exact recurrences and reuse	Restricted vacuum family, loops 1–12	543.8× faster

For the restricted vacuum family, exact recurrences express the coefficients through $z = L(d-2)$. This lets one calculation serve several loop orders and targets. The final method reuses intermediate coefficients, applies exact arithmetic and checks each result with an independent Taylor expansion. Appendix D derives both calculations. Kira remains available if the direct check fails. The measured gain includes these mathematical changes and program improvements; it does not isolate a machine learning effect.

Eight fresh groups each contain three mass profiles, with two, three or four massive denominators, and request loop orders one through twelve. Five repetitions compare the saved original Kira 3.1 executable, a later full-pool control and the recurrence method on identical requests. All 120 calculations pass exact checks. The final/original ratio is 0.001839, giving a speedup of 543.815 with interval $[488.134, 605.064]$. A separate check verifies 27,960 nonzero target–master

coefficient polynomials. The final method needs no Kira calls on these successful requests; each baseline method makes 1,440 calls across its trials.

This benchmark includes construction of Euler and mass partial-fraction identities. It does not generate a general high-loop IBP system. Earlier checks at loop orders 13, 16, 24 and 32 use the same restricted family, and the recurrence proof permits any positive integer loop order. General high-loop scattering remains outside this result.

11 Conclusion

The original seed and equation models save time in their stated test domains. We omit learned row selection because its complete-time comparisons give no reliable saving. Additional scattering training improves equation sufficiency, and the subsequent cost studies compare whole calculations with all retries included.

The new experiment fits seventy candidates and chooses a separate policy for each of seven graph structures. Validation activates a new cost model for 3 of the seven graphs and retains a control strategy for the others. The selected route gives 6.3% less time relative to native choices on fresh rational profiles, with time ratio 0.9368 with descriptive interval $[0.8867, 0.9878]$. Its comparison with a fixed strategy for each graph gives 0.9441 with descriptive interval $[0.8992, 0.9863]$. The aggregate interval lies below equal time. All 720 timing trials and 43 further applications pass exact checks. The per-graph measurements and controls show where learned selection, fixed settings, or fallback determine the result.

Sharing compatible requests gives a separate 48.3% saving. The 543.8-fold saving through twelve loops comes from exact recurrences and reuse in a restricted vacuum family. These results measure different calculations and do not define a combined product of speedups.

The native comparisons cover known structures at one through three loops. The newest test retains the training target patterns and changes their kinematic parameters. It does not establish a significant learned gain for every possible integral type. General high-loop scattering and globally optimal reduction paths remain outside the demonstrated results.

A IBP relations and exact elimination

Write $P_{\boldsymbol{\sigma}} = \prod_i D_i^{-\sigma_i}$ and $\partial_k^\mu = \partial/\partial \ell_{k\mu}$. Applying the product rule to the integrand in Eq. (2) gives

$$\partial_k^\mu (v_\mu P_{\boldsymbol{\sigma}}) = (\partial_k \cdot v) P_{\boldsymbol{\sigma}} + v \cdot \partial_k P_{\boldsymbol{\sigma}}, \tag{6}$$

$$v \cdot \partial_k P_{\boldsymbol{\sigma}} = -\sum_i \sigma_i (v \cdot \partial_k D_i) D_i^{-\sigma_i - 1} \prod_{j \neq i} D_j^{-\sigma_j}. \tag{7}$$

For $v = \ell_h$, the derivative gives $\partial_k \cdot v = d\delta_{kh}$, where δ_{kh} is one for $k = h$ and zero otherwise. For an external momentum, the derivative is zero. The integrated total derivative vanishes where the integral converges. Analytic continuation then defines the identity in dimensional regularization [2, 1]. This prescription also sets scaleless integrals to zero.

With a complete scalar-product basis, the scalar numerator $v \cdot \partial_k D_i$ can be written as $u_i + \sum_j v_{ij} D_j$. The coefficients u_i and v_{ij} depend on masses, invariants and the chosen momentum routing. Substituting in Eq. (7) gives

$$0 = \delta_v I_{\boldsymbol{\sigma}} - \sum_i \sigma_i u_i I_{\boldsymbol{\sigma} + \boldsymbol{e}_i} - \sum_{i,j} \sigma_i v_{ij} I_{\boldsymbol{\sigma} + \boldsymbol{e}_i - \boldsymbol{e}_j}, \tag{8}$$

where $\delta_v = \partial_k \cdot v$ and $\boldsymbol{e}_i$ increments index i by one. Thus differentiation produces linear relations with shifted integer indices. Auxiliary denominators or separately supplied algebraic relations

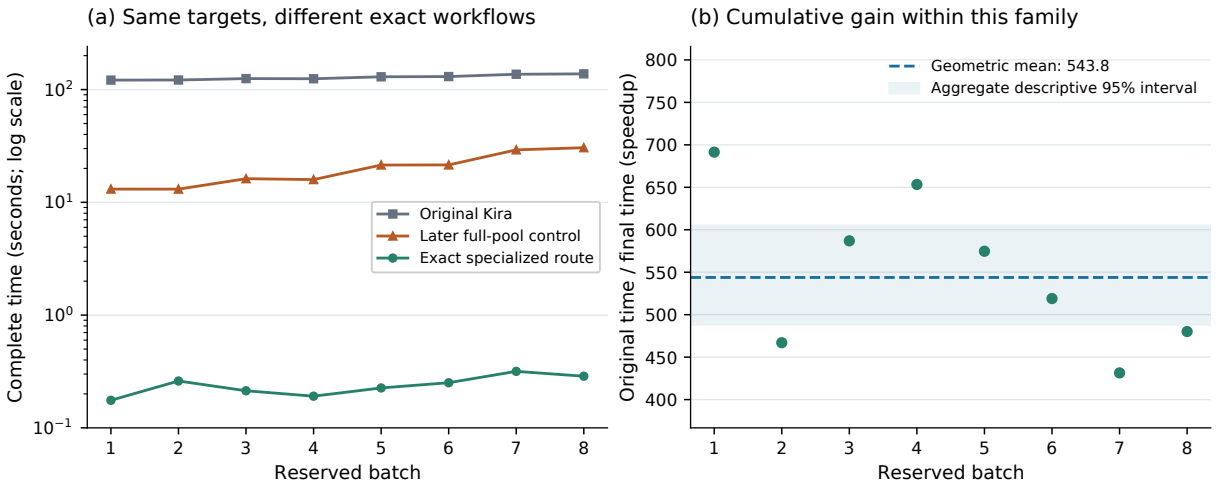


Figure 10: Exact recurrence reuse gives the large saving in the restricted vacuum family. Left: full times on a logarithmic axis. Right: original time divided by final time. Each batch contains three mass profiles at loops 1–12. The result combines mathematical reuse and program improvements.

handle dependent representations; the restricted family in Appendix D uses explicit mass partial fractions.

Suppose row q of such a system satisfies $\sum_j A_{qj} I_j = 0$ and $A_{qk} \neq 0$. Solving this row for the pivot integral gives

$$I_k = -\sum_{j \neq k} \frac{A_{qj}}{A_{qk}} I_j. \tag{9}$$

For a second identity row, or a target coefficient row r, subtract (A_{rk}/A_{qk}) times the zero identity. The resulting coefficients are

$$A'_{rj} = A_{rj} - \frac{A_{rk}}{A_{qk}} A_{qj}, \qquad A'_{rk} = 0. \tag{10}$$

Subtracting a zero identity leaves the target unchanged. A different allowed source row changes the intermediate expressions but preserves this equality. Repeating the operation removes the nonmaster columns and gives Eq. (3). Each pivot coefficient must be nonzero. Kinematic points at which a required denominator vanishes need a separate calculation.

B Reverse construction and the meaning of reference labels

For each checked source identity E_q, its coefficients sum to zero when evaluated as integrals. Substituting these identities into Eq. (4) gives

$$T - \sum_m r_m M_m = \sum_q \lambda_q E_q = 0. \tag{11}$$

This establishes the master coefficients of the constructed expression. Expanding each E_q in indexed integrals and collecting equal indices defines the input target. The reverse operations, source rows and their seed–operator labels supply a reproducible path back to the masters. The coefficients λ_q and r_m can be rational functions; constants form a special case. Mass partial-fraction identities have no IBP seed label.

Different zero identities can give different construction histories for the same target. A saved seed set is therefore sufficient, but need not be unique or smallest. Likewise, the saved row sequence need not use the fewest operations. A seed label records membership in one checked set, and a row label records the next step in one checked path. The final master coefficients and later steps do not enter the model inputs.

C Model specification and training objectives

The seed network needs numerical inputs for symbolic coefficient expressions. We split each expression into tokens and count individual tokens and consecutive pairs and triples. A fixed signed map places these counts in 64 components. Different token groups can share a component, so equal representations do not prove equal expressions. Integral indices, complexity counts and term positions give further inputs. A shared hyperbolic-tangent map converts each term into a vector $\boldsymbol{h}_u \in \mathbb{R}^{32}$. Family symbols, master indices and candidate indices give separate 32-component vectors.

The ordered-pair summary is

$$\boldsymbol{h}_{\mathrm{pair}} = \frac{1}{n_T(n_T-1)} \sum_{u \neq v} \psi_\theta([\boldsymbol{h}_u, \boldsymbol{h}_v, \boldsymbol{h}_u - \boldsymbol{h}_v, \boldsymbol{h}_u \odot \boldsymbol{h}_v]), \tag{12}$$

where n_T is the number of terms, square brackets concatenate vectors, $\odot$ multiplies components and ψ_θ returns 32 components. For $n_T = 1$, the summary is zero. This definition covers ordered pairs, not all larger term subsets. The candidate vector $\boldsymbol{u}_b$ weights target terms through scaled dot-product attention [9]:

$$w_{bu} = \frac{\exp(\boldsymbol{u}_b \cdot \boldsymbol{h}_u/\sqrt{32})}{\sum_v \exp(\boldsymbol{u}_b \cdot \boldsymbol{h}_v/\sqrt{32})}, \qquad \boldsymbol{v}_b = \sum_u w_{bu} \boldsymbol{h}_u. \tag{13}$$

The final network input joins the mean and maximum term vectors, pair vector, family vector, basis vector, seed vector and weighted-term vector. A 64-component hidden layer gives the output g_θ in Eq. (5). This construction uses shared maps and aggregation as in Ref. [8]. Stored positions retain term order. Reordering the candidate list only reorders the outputs.

If $y_b \in \{0, 1\}$ is recorded seed membership, a Bernoulli model assigns label likelihood $p_b^{y_b}(1 - p_b)^{1-y_b}$. Taking its negative logarithm gives $-y_b \log p_b - (1 - y_b) \log(1 - p_b)$. Averaging within each candidate list and then across targets gives the implemented loss

$$\mathcal{L}_s = -\frac{1}{|\mathcal{D}_s|} \sum_{T \in \mathcal{D}_s} \frac{1}{|\mathcal{C}(T)|} \sum_{b \in \mathcal{C}(T)} [y_b \log p_b + (1 - y_b) \log(1 - p_b)]. \tag{14}$$

Here $\mathcal{D}_s$ is the training-target set. This objective treats each target equally and does not claim that true seed dependencies are statistically independent. Training uses AdamW [10] with learning rate 10^{-3}, weight decay 10^{-4}, gradient norm limit one and up to 32 targets per update. The threshold 1/2 stays fixed during the expanded label test.

For row selection, let $\mathcal{A}$ be the current allowed action list and $\boldsymbol{f}_a \in \mathbb{R}^{132}$ its feature vector for action a. The encoder is $\boldsymbol{h}_a = \tanh(W\boldsymbol{f}_a + \boldsymbol{b})$ with 32 components. Concatenating $\boldsymbol{h}_a$ with the mean and componentwise maximum over $\mathcal{A}$ gives the input to a 64-component hidden layer. Its scalar output g_a defines

$$p(a \mid A, \mathcal{A}) = \frac{e^{g_a}}{\sum_{b \in \mathcal{A}} e^{g_b}}, \qquad \mathcal{L}_r = -\frac{1}{|\mathcal{D}_r|} \sum_{(A, \mathcal{A}, a^*) \in \mathcal{D}_r} \log p(a^* \mid A, \mathcal{A}). \tag{15}$$

Here A is the matrix, a^* is the saved action, and $\mathcal{D}_r$ is the set of training states with a choice. The denominator makes the probabilities sum to one. The negative logarithm of the reference action's probability gives the loss. A state with one action has zero loss. Training uses the seed model's AdamW settings and 256 states per update. Trial updates modulo a prime provide cancellation features. The numerical size of a modular value does not measure the size of a real coefficient.

Later row models learn the cost of an action. One fit predicts the logarithm of the measured insertion time. Another counts the remaining forward insertions after that action, with Kira choosing later steps. If insertion j uses n_j source entries, its estimated cost is $1 + 2n_j$: one

division, followed by a multiplication and subtraction for each entry. Thus the cost label is $C(S,q)=\sum_j(1+2n_j)$ for state S and choice q. The later tree fits

$$y_q=\frac{C(S,q)-C(S,0)}{\max\{1,\max_b C(S,b)-\min_b C(S,b)\}},\qquad \widehat{y}_q=h(\boldsymbol{x}_q-\boldsymbol{x}_0), \tag{16}$$

where 0 denotes Kira's native source and $\boldsymbol{x}_q$ contains eighteen current-row and overlap features. Weighted squared error gives each training family equal weight. Validation selects the tree and the required predicted improvement, and exact coefficients must agree before a measured continuation enters training. This count describes coefficient operations, while complete timings include the algebraic work inside each coefficient operation. Neither training objective proves a globally minimum path.

The support of an equation is the set of integrals with nonzero coefficients. An equation label records whether this support appears in Kira's selected system. Table 11 groups the equation tree's seventeen inputs. They describe the seed and target before symbolic coefficient construction. For seed indices $\boldsymbol{\sigma}$ and target indices $\boldsymbol{a}$, the distance is $\sum_i|\sigma_i-a_i|$. The additional-power count is $\sum_i\max(\sigma_i-1,0)$, and the numerator rank is $\sum_i\max(-\sigma_i,0)$. Neither quantity requires equation coefficients.

Table 11: Inputs to the original equation tree. The scattering seed trees omit the operator input and use sixteen features.

Group	Quantities	Number
Family and request	Loops, external momenta, index count, logarithm of target count	4
Seed	Positive indices, additional powers, numerator rank	3
Target limits	Excess seed powers and rank above target maxima	2
Sector	Whether requested; excess powers and rank above its target maxima	3
Target distance	Minimum index distance; whether the seed is a target	2
Operator	IBP operator index	1
Overall limits	Maximum target powers and numerator rank	2

Training gives each physical profile equal total weight by assigning each row weight $1/(n_bB_g)$, where n_b is its batch size and B_g is the number of training batches in its profile. At a node, let w_+,w_- be the positive and negative total weights. Half the weighted binary Gini impurity is

$$H=\frac{w_+w_-}{w_++w_-}. \tag{17}$$

Indeed, with $p=w_+/(w_++w_-)$, the weighted Gini expression is $(w_++w_-)[1-p^2-(1-p)^2]=2H$. Each proposed split reduces H by the sum of its two children's values. The fit tests 25 feature quantiles between 0.02 and 0.98, rejects children below the stated population limit, and selects the largest positive reduction. This is the classification-tree construction of Ref. [12] with our bounded set of candidate splits.

A leaf with n_+ positive rows among n rows returns

$$p_{\mathrm{leaf}}=\frac{n_++1/2}{n+1}. \tag{18}$$

Adding one half-count to each class gives a finite score even when a leaf contains only one class. The original tree keeps an equation when $p_{\mathrm{leaf}}\geq 0.01$. These scores estimate reference-set membership; they are not calibrated probabilities that the whole reduction succeeds. The depth-eight and depth-twelve fits require at least 24 and eight rows in each child. The split procedure follows Ref. [12]. A basis check and fallback still test whether the selected system is sufficient.

C.1 Additional scattering seed and equation trees

The added scattering study fits a separate pair of trees for each of the three graph structures. The equation labels use the same selected-support rule as the original equation model. Let $y_{\sigma,o}$ be one when the native selected system contains the recorded equation support for seed σ and operator o. A seed receives label one when any of its operators has label one:

$$y_\sigma = \max_o y_{\sigma,o}. \tag{19}$$

For binary labels, this maximum is one exactly when at least one selected-support equation comes from the seed. This proves the stated membership rule. The label gives a useful reference set, not a proof that every labelled seed is necessary.

Each seed appears once per training batch. Its tree uses the sixteen features that do not depend on the operator. Thus every operator of that seed receives the same score. The early filter either permits or skips construction of all its equations. Kira still enumerates the seed region and applies its native symmetries. The equation tree uses all seventeen features and can keep different operators of the same seed.

Both fits have maximum depth twelve and at least eight records in each child. Each training batch has equal total weight. The split rule and leaf score are those derived above. Validation compares thresholds 0.001, 0.01 and 0.05 using full reduction time. Three repetitions contribute to each validation median. The chosen threshold minimizes the geometric mean ratio over the rational and symbolic validation batches of its graph. Model parameters and thresholds are fixed before testing. For the nonplanar box, both thresholds are 0.05. For the pentagon, the seed and equation thresholds are 0.05 and 0.001. For the box with virtuality, they are 0.001 and 0.01. These fits create new small tree models; they do not change the original neural checkpoints. The reserved timing test gives no overall saving, so the application keeps native choices as its default.

D Exact reuse for the restricted vacuum family

The cumulative speedup concerns the following Euclidean vacuum family. Its squared masses $\mu_j = \rho_j \mu$ are positive and distinct:

$$V_{\boldsymbol{b}}^{(L)} = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{\pi^{d/2}} \frac{1}{\ell_1^2 \cdots \ell_L^2} \prod_{j=1}^{R} (Q^2 + \mu_j)^{-b_j}, \qquad Q = \sum_{h=1}^{L} \ell_h. \tag{20}$$

Here b_j is a nonnegative integer, $B = \sum_j b_j \geq 1$, R is the number of massive denominators, μ is a common squared-mass scale and ρ_j is a positive mass ratio. Each massless denominator has power one. All massive denominators share Q, so this family has special algebraic dependencies. Its prescribed masters are $M_j^{(L)} = V_{\boldsymbol{e}_j}^{(L)}$. The comparison uses $R = 2, 3, 4$ and $\mu = 1$ in the selected units.

Let f be the integrand without the measure. The simultaneous Euler derivative gives $\sum_h \ell_h \cdot \partial_{\ell_h} (\ell_i^2)^{-1} = -2(\ell_i^2)^{-1}$ for each massless factor. Also, $\sum_h \ell_h \cdot \partial_{\ell_h} Q^2 = 2Q^2$. Therefore

$$\sum_h \partial_{\ell_h} \cdot (\ell_h f) = \left[Ld - 2L - 2 \sum_j b_j \frac{Q^2}{Q^2 + \mu_j} \right] f. \tag{21}$$

Use $Q^2 = (Q^2 + \mu_j) - \mu_j$ and integrate the derivative as in Appendix A. This gives

$$0 = [L(d-2) - 2B] V_{\boldsymbol{b}}^{(L)} + 2 \sum_j b_j \mu_j V_{\boldsymbol{b}+\boldsymbol{e}_j}^{(L)}. \tag{22}$$

For two present massive denominators, the integrand identity $(Q^2+\mu_i)-(Q^2+\mu_j)=\mu_i-\mu_j$ separately gives

$$(\mu_i-\mu_j)V^{(L)}_{\boldsymbol{b}}=V^{(L)}_{\boldsymbol{b}-\boldsymbol{e}_i}-V^{(L)}_{\boldsymbol{b}-\boldsymbol{e}_j}. \tag{23}$$

This second relation is an algebraic partial fraction, so it introduces no additional IBP seed.

Define $z=L(d-2)$ and $W^{(L)}_{\boldsymbol{b}}=\mu^{B-1}V^{(L)}_{\boldsymbol{b}}$. The factor of μ gives W the masters' mass dimension. Substituting in Eqs. (22) and (23) removes the common scale. If $\boldsymbol{C}_{\boldsymbol{b}}(z)$ is the vector of master coefficients of $W^{(L)}_{\boldsymbol{b}}$, then

$$\boldsymbol{C}_{\boldsymbol{b}}(z)=\frac{\boldsymbol{C}_{\boldsymbol{b}-\boldsymbol{e}_i}(z)-\boldsymbol{C}_{\boldsymbol{b}-\boldsymbol{e}_j}(z)}{\rho_i-\rho_j}, \qquad b_i,b_j>0, \tag{24}$$

$$\boldsymbol{C}_{k\boldsymbol{e}_j}(z)=\frac{2(k-1)-z}{2(k-1)\rho_j}\boldsymbol{C}_{(k-1)\boldsymbol{e}_j}(z), \qquad k\geq 2. \tag{25}$$

The initial vector is $\boldsymbol{C}_{\boldsymbol{e}_j}=\boldsymbol{e}_j$. Every recurrence lowers B, so it terminates at the prescribed masters. Its coefficients depend on L only through z. Induction on B shows that they are polynomials in z with rational coefficients at fixed rational ρ_j. One calculation can therefore supply all requested loop orders by substituting $z=L(d-2)$.

We independently check the mixed-denominator reduction through Taylor partial fractions. For a dimensionless auxiliary variable x, write

$$\prod_i (x+\rho_i)^{-b_i}=\sum_{j:b_j>0}\sum_{k=1}^{b_j}\frac{a_{jk}}{(x+\rho_j)^k}. \tag{26}$$

At the pole $x=-\rho_j$, set $t=x+\rho_j$. Multiplying by t^{b_j} and expanding the regular factors shows that the principal-part coefficient is

$$a_{jk}=[t^{b_j-k}]\prod_{i\neq j}(\rho_i-\rho_j+t)^{-b_i}. \tag{27}$$

Here $[t^n]$ extracts Taylor coefficient n and factors with $b_i=0$ are one. Applying Eq. (25) repeatedly to each pole term gives

$$C_{\boldsymbol{b}j}(z)=\sum_{k=1}^{b_j}a_{jk}\prod_{h=1}^{k-1}\frac{2h-z}{2h\rho_j}, \qquad V^{(L)}_{\boldsymbol{b}}=\mu^{1-B}\sum_j C_{\boldsymbol{b}j}(L(d-2))M^{(L)}_j. \tag{28}$$

An empty product is one. The Taylor calculation avoids the selected pair-recurrence path and checks every returned polynomial coefficient before substitution. The derivation assumes distinct ρ_j, unit massless powers and the shared momentum Q; it supplies no formula for arbitrary multiloop scattering families.

E Measurement definitions and comparison conditions

For seed labels, let TP count selected reference seeds, FP count selected seeds absent from the reference set, and FN count omitted reference seeds. Precision is $P=\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$ and recall is $R=\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$. Defining F_1 as their harmonic mean gives

$$F_1=\frac{2}{P^{-1}+R^{-1}}=\frac{2PR}{P+R}=\frac{2\mathrm{TP}}{2\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}. \tag{29}$$

Set agreement instead counts targets whose entire predicted set equals the recorded set. Algebraic sufficiency counts targets that reduce to the prescribed basis with the selected source equations. These definitions explain why the three quantities in Figure 1 can differ.

For timing, let T_{abr} be the measured time for method a, batch b and repetition r. We use $\widetilde{T}_{ab} = \mathrm{median}_{r=1,\dots,5} T_{abr}$. For the later native comparisons, if physical group g contains batches B_g, define the equal-group mean logarithmic ratio

$$\bar{\ell} = \frac{1}{G}\sum_{g=1}^{G}\frac{1}{|B_g|}\sum_{b\in B_g}\log\frac{\widetilde{T}_{ab}}{\widetilde{T}_{cb}}, \qquad R_{a/c} = e^{\bar{\ell}}. \tag{30}$$

Here c identifies the control and G is the number of physical groups. Exponentiating the mean log ratio gives the geometric mean; ratios below one indicate lower time. Reversing numerator and denominator changes $\bar{\ell}$ to $-\bar{\ell}$, so speedup is $S = 1/R_{a/c}$ and the time reduction is $100(1-R_{a/c})\%$. The cumulative study uses one batch per group. The supplied-equation seed comparison instead gives each complete paired batch equal weight; its interval resamples six related kinematic groups. Following bootstrap resampling [14], reported descriptive 95% intervals resample complete groups 10,000 times while retaining paired methods and repetitions. They describe case-to-case variation in these experiments. We do not infer new uncertainty intervals for figures whose source summaries provide only point estimates.

The supplied-equation seed test includes target preparation, inference, early basis checking, reduction, result checking and fallback. It excludes original IBP-pool generation. It rotates methods within each family and target batch on the same node, and includes only complete paired batches in its ratio summary while preserving incomplete and skipped trials in the accounting.

The native equation and transfer tests include input preparation, model loading and evaluation, Kira startup, native IBP generation, elimination, basis and coefficient checks, cleanup and every fallback. Independent reference discovery and the later independent check lie outside the timer. In the equation test, the native and learned arms receive the same executable-level improvements. The eight reserved profiles test new masses and invariants within known structures; the seven transfer cases additionally test new structures and symbolic invariants.

The cumulative test includes fresh-worker startup, input loading, supplied Euler and mass-identity construction, algebra, verification, output saving and complete output reading. Eight groups use maximum massive powers 8, 10, 12 or 14, with two groups at each bound. They request 16, 32, 48 or 64 targets per profile, capped by availability. Each group contains three mass profiles and all loop orders 1–12. Paired methods use the same resources, and their order changes between repetition blocks. Each method recomputes the reduction. Waiting for resources, offline training and the later independent check remain outside all reported production times.

The initial live-matrix comparison uses three repetitions of four methods on seven physical cases, giving 84 workflows. Its complete ratio is the geometric mean of the seven paired median ratios and carries no reported uncertainty interval. The later native implementation comparison uses five repetitions and sixteen batches; it repeats earlier physical profiles to measure the implementation change.

The reported numbers are measurements from this study. We keep the original label tests, seed tests, row tests, equation tests, scattering tests and vacuum benchmark separate. This prevents differences in targets or timed work from being mistaken for a combined model gain. The archived notes preserve earlier experiments, including negative results.

F Integral families and kinematic conventions

To specify a family in Eq. (1), we give its ordered denominator momenta, masses and external scalar products. The definitions below cover the training families and the reported test families. A target is an index vector in one family or a rational linear combination of such vectors. These definitions specify the generated integrals without listing millions of individual records. Unless stated otherwise, $D_i = q_i^2 - m_i^2 + i0$ and the measure is that of Eq. (1). The Euclidean vacuum family keeps the sign convention in Eq. (20).

F.1 Initial one-loop and two-loop families

For the one-loop tadpole, $q_1 = \ell$ and $m_1^2 = \mu > 0$. For the bubbles,

$$\boldsymbol{q} = (\ell, \ell - p), \qquad p^2 = s, \qquad (m_1^2, m_2^2) \in \{(0,0), (\mu,0), (\mu,\mu), (\mu_1,\mu_2)\}. \tag{31}$$

The symbols μ, μ_1, μ_2 denote squared masses, with $\mu_1 \neq \mu_2$ in the unequal-mass case. The triangle routing is

$$\boldsymbol{q} = (\ell, \ell - p_1, \ell - p_1 - p_2), \qquad (p_1^2, p_2^2, (p_1+p_2)^2) = (u_1, u_2, u_3). \tag{32}$$

Expanding $(p_1+p_2)^2$ gives $2p_1 \cdot p_2 = u_3 - u_1 - u_2$. Three massless triangle classes use virtualities $(0,0,s)$, $(0,u_2,u_3)$ and (u_1,u_2,u_3). Two equal-mass classes, with all internal squared masses μ, use $(0,0,s)$ and (u_1,u_2,u_3). Each displayed nonzero virtuality is independent at generic kinematics.

The initial massless boxes use four incoming external momenta, $p_4 = -p_1 - p_2 - p_3$, and

$$\boldsymbol{q} = (\ell, \ell - p_1, \ell - p_1 - p_2, \ell - p_1 - p_2 - p_3), \quad s = (p_1+p_2)^2, \quad t = (p_2+p_3)^2. \tag{33}$$

Writing $u_i = p_i^2$, the three classes have $(u_1,u_2,u_3,u_4) = (0,0,0,0)$, $(0,0,0,u_4)$ and (u_1,u_2,u_3,u_4). Expanding the definitions of s, t and p_4^2 gives

$$2p_1 \cdot p_2 = s - u_1 - u_2, \quad 2p_2 \cdot p_3 = t - u_2 - u_3, \quad 2p_1 \cdot p_3 = u_4 + u_2 - s - t. \tag{34}$$

These definitions give thirteen one-loop classes. The initial two-loop massless two-point family uses

$$\boldsymbol{q} = (\ell_1, \ell_2, \ell_1 - \ell_2, \ell_1 - p, \ell_2 - p), \qquad p^2 = s, \qquad m_i^2 = 0. \tag{35}$$

The equal-mass sunrise uses $\boldsymbol{q} = (\ell_1, \ell_2, p - \ell_1 - \ell_2, p - \ell_1, p - \ell_2)$, with masses $(\mu,\mu,\mu,0,0)$ and $p^2 = s$. Its last two entries are auxiliary: $a_4, a_5 \leq 0$. The equal-mass vacuum sunset has $\boldsymbol{q} = (\ell_1, \ell_2, \ell_1 - \ell_2)$ and all three squared masses μ. These three classes complete the initial sixteen. Their subsectors include the integrals obtained by removing physical denominators.

F.2 Additional one-loop mass and invariant assignments

The 984 added box families use a more general assignment than the three initial massless boxes:

$$D_i = (\ell - Q_i)^2 - \mu_{\alpha_i} + i0, \quad Q_1 = 0, \quad \mu_0 = 0, \quad \alpha_i \in \{0,1,2,3\}. \tag{36}$$

The nonzero squared masses μ_1, μ_2, μ_3 are independent. Repeated labels impose equal masses. The three external shifts Q_2, Q_3, Q_4 have squared differences

$$\Delta_{ij} = (Q_i - Q_j)^2 = y_{\beta_{ij}}, \quad y_0 = 0, \quad \beta_{ij} \in \{0,1,2,3\}. \tag{37}$$

The nonzero y_1, y_2, y_3 are independent invariants. Expanding a squared difference gives the Gram matrix

$$Q_i \cdot Q_j = \frac{\Delta_{1i} + \Delta_{1j} - \Delta_{ij}}{2}, \qquad i,j \in \{2,3,4\}. \tag{38}$$

The construction rejects singular sampled Gram matrices and identifies assignments related by propagator permutations or by relabelling nonzero masses and invariants. A fixed order selects 984 of 1,673 admissible inequivalent profiles after excluding the original massless boxes. This extends the initial collection to 1,000 families with 9,678 family-local masters and 240,472 targets before the higher-loop additions. Each profile gives a symbolic family through its zeros and equality conditions. The single-target seed timing comparison uses twelve reserved instances of these boxes. Its targets follow the reverse construction in Appendix B, rather than the native target enumeration below.

F.3 Native multiloop comparisons

The sunrise comparisons allow independent masses on the same three physical propagators. The other native comparison families use

$$\begin{aligned}\boldsymbol{q}_{\mathrm{DB}} &= (\ell_1, \ell_1 - p_1, \ell_1 - p_1 - p_2, \ell_2, \ell_2 - p_1 - p_2, \ell_2 - p_1 - p_2 - p_3,\\ &\qquad \ell_1 - \ell_2, \ell_1 - p_3, \ell_2 - p_1), \qquad (39)\\ \boldsymbol{q}_{\mathrm{V}} &= (\ell_1, \ell_2, \ell_3, \ell_1 - \ell_2, \ell_2 - \ell_3, \ell_1 - \ell_3), \qquad (40)\\ \boldsymbol{q}_{\mathrm{L}} &= (\ell_1, \ell_1 - p, \ell_2, \ell_2 - p, \ell_3, \ell_3 - p, \ell_1 - \ell_2, \ell_2 - \ell_3, \ell_1 - \ell_3). \qquad (41)\end{aligned}$$

The double box has two loops and seven physical denominators; its last two entries are auxiliary. The vacuum has three loops and six physical denominators. The ladder has three loops and eight physical denominators; its last entry is auxiliary. Auxiliary indices remain nonpositive. The massless double box has $p_i^2 = 0$ and uses $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = t$, $2p_2 \cdot p_3 = -s - t$. This local definition of t differs from the initial box convention because the routing differs.

For the final early-equation test, the sunrise squared-mass vectors are $(2, 7, 11)$ and $(3, 7, 13)$, with $p^2 = -101, -103$. The double-box profiles are $(s, t) = (-101, -103), (-107, -109)$. The vacuum squared-mass vectors are $(1, 1, 1, 1, 1, 11)$ and $(1, 1, 1, 13, 13, 13)$. The ladder has squared masses $(1, 0, 0, 0, 0, 0, 0, 0)$ and $p^2 = -101, -103$. All values use a common squared-momentum unit. These choices keep d symbolic and give coefficients in $\mathbb{Q}(d)$.

The target order starts with the full-sector corner, then adds each single extra propagator power and each available single auxiliary numerator. The construction repeats after removing one, two and three physical denominators, retains at least two, and removes duplicates. The comparison takes 21 sunrise targets and batches of 32 and 128 targets for each other profile. This gives the fourteen batches in Section 6.

F.4 New scattering structures and symbolic invariants

The massless nonplanar box uses the routing of Ref. [4]:

$$\begin{aligned}\boldsymbol{q}_{\mathrm{NP}} = (&\ell_1, \ell_2, p_1 - \ell_1, p_2 - \ell_2, p_1 + p_2 - p_3 - \ell_1,\\ &p_1 - \ell_1 + \ell_2, p_1 - p_3 - \ell_1 + \ell_2, \ell_1 - p_3, \ell_2 - p_1 - p_2). \qquad (42)\end{aligned}$$

Its first seven entries are physical. Its last two are auxiliary. Here $p_1^2 = p_2^2 = p_3^2 = (p_1 + p_2 - p_3)^2 = 0$, $s = (p_1 + p_2)^2$ and $t = (p_1 - p_3)^2$. Expanding these squares gives $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = -t$ and $2p_2 \cdot p_3 = s + t$.

The massless pentagon uses $\sum_{i=1}^{5} p_i = 0$, $p_i^2 = 0$ and

$$\boldsymbol{q}_{\mathrm{P}} = (\ell, \ell - p_1, \ell - p_1 - p_2, \ell - p_1 - p_2 - p_3, \ell - p_1 - p_2 - p_3 - p_4). \qquad (43)$$

All five denominators are physical. The independent invariants are $(s_{12}, s_{23}, s_{34}, s_{45}, s_{51})$, where $s_{ij} = (p_i + p_j)^2$. Squaring $p_1 + p_2 + p_3 = -(p_4 + p_5)$ gives $2p_1 \cdot p_3 = s_{45} - s_{12} - s_{23}$; the other nonadjacent products follow by cyclic relabelling. The box with an external virtuality uses $\boldsymbol{q} = (\ell, \ell - p_1, \ell - p_1 - p_2, \ell - p_3)$, $p_1^2 = p_2^2 = p_3^2 = 0$, and $v = (p_1 + p_2 - p_3)^2$. With the nonplanar convention for s, t, expanding v gives $2p_2 \cdot p_3 = s + t - v$.

The rational nonplanar profiles are $(s, t) = (-149, -147), (-167, -157)$. The pentagon profiles are $(-51, -53, -57, -59, -63)$ and $(-71, -73, -77, -79, -83)$ in the displayed invariant order. Each rational profile supplies sixteen targets. The symbolic nonplanar, pentagon and virtuality-box tests supply ten, six and eight targets. They start with the corner and single additional powers, followed by auxiliary numerators and single removed denominators where available. These definitions give the 88 targets of the transfer test.

F.5 Parameter points and targets for the additional training

The new scattering study uses the same three denominator lists and scalar-product conventions defined above. For integer profile number n, we choose

$$(s,t)_{\mathrm{NP}} = (-113-18n, -127-10n), \tag{44}$$

$$(s_{12}, s_{23}, s_{34}, s_{45}, s_{51})_{\mathrm{P}} = (-11-20n, -13-20n, -17-20n, -19-20n, -23-20n), \tag{45}$$

$$(s,t,v)_{\mathrm{box}} = (-113-18n, -127-10n, -149-12n). \tag{46}$$

These are specified input values in a common squared-momentum unit. Training uses $n = 10, 11$, validation uses $n = 12$, and testing uses $n = 13, 14$. The dimension d remains symbolic in every case. Each symbolic batch instead keeps all of its displayed invariants independent.

The target list starts with the full-sector corner. It then raises each physical denominator power once and adds each single auxiliary numerator. The same sequence follows for each possible single removed physical denominator. Rational batches take the first sixteen entries. Symbolic training takes entries 0–3, validation takes 4–7, and testing takes 8–11, with indices starting at zero. We verify that the three symbolic lists are disjoint for each structure. Rational batches reuse index templates at different parameter points, so this split does not claim entirely unseen target patterns in every representation.

G Master bases and dimensionally consistent target construction

For a complete independent family basis, the number of scalar products involving loop momenta is

$$N_{\mathrm{sp}} = \frac{L(L+1)}{2} + LE. \tag{47}$$

There are L loop squares and $L(L-1)/2$ distinct loop cross-products, followed by LE products with the E independent external momenta. This count supplies a direct rank check for the independent denominator lists. The restricted vacuum parent in Eq. (20) instead uses the explicitly dependent massive denominators and their partial fractions.

For index vector $\boldsymbol{a}$, define the positive-index sum $r = \sum_i \max(a_i, 0)$, numerator rank $s_- = \sum_i \max(-a_i, 0)$ and additional-power count $b = \sum_i \max(a_i - 1, 0)$. If the top sector has t_{top} physical denominators, the initial Kira basis calculation uses seed limits $(r, s_-, b) \le (t_{\mathrm{top}}+2, 2, 2)$ and probe limits $(t_{\mathrm{top}}+1, 1, 1)$, component by component. Both include all subsectors. The basis check then increases every limit by one and repeats at a second finite-field point. The original sixteen definitions give the same master labels in all three calculations, for 48 completed basis checks. Such agreement supports stability in the tested domains; it does not prove completeness for all integer indices [4, 5].

The initial sixteen classes give 49 family-local master entries. The tadpole gives one. The four bubble classes give one, two, two and three, in their stated order. The five triangle classes give one, two, four, three and five. The three box classes give three, four and eleven. The massless two-loop two-point, sunrise and vacuum sunset classes give two, three and two. For example, the equal-mass sunrise basis is

$$M_1 = I_{(1,1,0,0,0)}, \qquad M_2 = I_{(1,1,1,0,0)}, \qquad M_3 = I_{(1,1,1,-2,0)}. \tag{48}$$

The first integral separates into two tadpoles, and the last contains the numerator $[(p-\ell_1)^2]^2$. These are basis integrals, whose numerical values need not be calculated to train a reduction policy. Each later family instance retains its own explicit index basis and parameter assignment in the integral data.

The mass dimension of Eq. (1) is $Ld-2\sum_i a_i$: each loop measure contributes d powers of mass and each inverse propagator contributes two. In the restricted vacuum data, $[V_{\boldsymbol{b}}^{(L)}] = Ld-2L-2B$. Thus targets of different total powers can be combined consistently as

$$T = \sum_{\nu} \kappa_\nu d^{\eta_\nu} \mu^{B_\nu - B_*} V_{\boldsymbol{b}_\nu}^{(L)}, \qquad B_* = \max_\nu B_\nu. \tag{49}$$

Every term has dimension $Ld-2L-2B_*$. The construction uses $\eta_\nu \in \{0,1\}$ and $\kappa_\nu = n_\nu/q_\nu$, with $n_\nu \in \{-3,-2,-1,1,2,3\}$ and $q_\nu \in \{1,2,3\}$. Each expression contains one, two or three distinct roots. Individual roots have $2 \le B \le 5$; Euler seeds have $1 \le B \le 5$, and mass identities extend through $B = 6$. Each added family contains 128 distinct targets, including individual roots. The identity pools contain 35, 160 or 545 rows for two, three or four massive denominators.

For integer profile number n, the generated mass ratios are $R = 2 + ((n-1) \bmod 3)$, $\rho_1 = 1$ and $\rho_{j+2} = 2 + 3n + j(n+2)$ for $0 \le j \le R-2$. Profiles $n = 1, \ldots, 100$ enter at loop orders 2–12; profiles 101–1350 enter at orders 1–12. Distinct masses make the partial-fraction denominators nonzero. A profile remains in the same training or testing group at every loop order. These rules state the extent of the high-loop collection and its restricted algebraic structure.

G.1 Analytic check of the restricted masters

The recurrences in Appendix D reduce the specified family to $M_j^{(L)}$. We can also derive their values to check this choice. Write $\delta = d/2$, $\alpha = L(\delta-1)$ and $c = L-(L-1)\delta$. The symbol Γ denotes the Euler gamma function, and a, b below are denominator powers. The standard Euclidean two-denominator convolution follows from Feynman parameters [1]:

$$\begin{aligned} J(a,b;Q) &= \int \frac{d^dk}{\pi^\delta} \frac{1}{(k^2)^a((Q-k)^2)^b} \\ &= (Q^2)^{\delta-a-b} \frac{\Gamma(a+b-\delta)\Gamma(\delta-a)\Gamma(\delta-b)}{\Gamma(a)\Gamma(b)\Gamma(2\delta-a-b)}. \end{aligned} \tag{50}$$

To obtain this expression, combine the denominators with

$$\frac{1}{A^a B^b} = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 \frac{x^{a-1}(1-x)^{b-1}\,dx}{[xA+(1-x)B]^{a+b}}. \tag{51}$$

Shifting k completes the square and gives the radial denominator $k^2 + x(1-x)Q^2$. The radial integral equals $\Gamma(a+b-\delta)[x(1-x)Q^2]^{\delta-a-b}/\Gamma(a+b)$. The remaining beta integral is $\Gamma(\delta-b)\Gamma(\delta-a)/\Gamma(2\delta-a-b)$, which yields Eq. (50).

At fixed Q, successive convolution of the L unit-power massless factors gives

$$C_L (Q^2)^{-c}, \qquad C_L = \frac{\Gamma(c)\Gamma(\delta-1)^L}{\Gamma(\alpha)}. \tag{52}$$

For $L = 1$ the coefficient is one. Applying Eq. (50) with $a = c$, $b = 1$ advances c to $c+1-\delta$ and proves the formula by induction. The final integral over Q then gives

$$\begin{aligned} M_j^{(L)} &= \frac{C_L}{\Gamma(\delta)} \int_0^\infty \frac{x^{\alpha-1}\,dx}{x+\mu_j} \\ &= \mu_j^{\alpha-1} \frac{\Gamma(c)\Gamma(\delta-1)^L\Gamma(1-\alpha)}{\Gamma(\delta)}. \end{aligned} \tag{53}$$

The change $x = \mu_j y$ reduces the integral to $\Gamma(\alpha)\Gamma(1-\alpha)$ for $0 < \operatorname{Re}\alpha < 1$. Analytic continuation gives the dimensionally regulated result [1]. Distinct positive fixed mass ratios give distinct exponentials $\rho_j^{L(d/2-1)-1}$ as functions of d. After common factors cancel, a rational-function

relation would have a unique largest exponential as real d increases. Rational coefficients grow at most as powers and cannot cancel that term. Removing its zero coefficient and repeating establishes independence over $\mathbb{Q}(d)$ at these fixed mass ratios. Special dimensions and coincident masses require their own analysis.

H Finite minimum-operation checks

A saved construction proves that a reduction path exists. To prove that a path has minimum cost, we must also specify the allowed equations, operations and cost. We perform this check for two small systems. Define the massless bubble $B_{a,b} = I_{(a,b)}$ with denominators $\ell^2 + i0$ and $(\ell-p)^2+i0$, where $p^2 = s \neq 0$. Contracting the derivative with ℓ uses $2\ell\cdot(\ell-p) = \ell^2+(\ell-p)^2-s$. Equation (7) then gives

$$0 = (d-2a-b)B_{a,b} - bB_{a-1,b+1} + bsB_{a,b+1}. \tag{54}$$

Changing variables $\ell \mapsto p-\ell$ exchanges a, b and gives the second operator identity. Integrals with only a massless denominator vanish in dimensional regularization [1]. The two operators at $(1,1)$, the first at $(2,1)$ and the second at $(1,2)$ therefore give

$$\begin{pmatrix} 0 & s & 0 & d-3 \\ s & 0 & 0 & d-3 \\ d-5 & -1 & s & 0 \\ -1 & d-5 & s & 0 \end{pmatrix} \begin{pmatrix} B_{2,1} \\ B_{1,2} \\ B_{2,2} \\ B_{1,1} \end{pmatrix} = 0. \tag{55}$$

The first two rows yield $B_{2,1} = B_{1,2} = (3-d)B_{1,1}/s$. Substituting in the third yields $B_{2,2} = (d-6)(d-3)B_{1,1}/s^2$.

For the massive tadpole $T_a = \int d^d\ell\,(\ell^2-m^2+i0)^{-a}/(i\pi^{d/2})$, the same derivative gives

$$0 = dT_a - 2a\int \frac{d^d\ell}{i\pi^{d/2}} \frac{\ell^2}{(\ell^2-m^2+i0)^{a+1}} = (d-2a)T_a - 2am^2T_{a+1}. \tag{56}$$

Seeds $a = 1, 2, 3$ reduce T_2, T_3, T_4 to T_1. We keep the master column protected in both systems. An allowed step selects a nonzero pivot in an unprocessed nonmaster column, normalizes its row and eliminates that column from the other rows. The cost counts one division per nonzero nonpivot source entry when normalization is needed, plus one multiplication and one subtraction for each nonpivot source-entry update. It excludes assignments and the internal polynomial operations of a rational-function operation.

Let S be the exact matrix state, $\mathcal{A}(S)$ its allowed actions, $c(S,a)$ the stated step cost and S_a the resulting state. Splitting every complete path into its first action and its remaining path gives the finite recurrence [15]

$$V(S) = \min_{a\in\mathcal{A}(S)} \{c(S,a) + V(S_a)\}, \qquad V(S_{\text{done}}) = 0. \tag{57}$$

Each action removes an unprocessed column. Induction on the number of remaining columns proves the minimum after all actions have been evaluated. Repeating this calculation for every source-row subset of nonmaster rank three also minimizes over initial equation selection.

Our bubble check covers 113 states, 234 transitions and 152 complete paths across five sufficient subsets. Its minimum is seven coefficient operations with subset selection and thirteen with the full pool. The tadpole check covers 24 states, 32 transitions and fifteen paths; its minimum is seven operations with its single sufficient subset. Each pool requires at least three equations. The smallest number of distinct seed labels is two for the bubble and three for the tadpole. Exact replay and six native Kira master-coefficient comparisons agree. These are minima for the displayed identities, protected basis, allowed operations and cost. They do not establish minimum elapsed time or an optimum over all IBP identities.

I Learning the cost of a complete reduction

The cost model receives a vector $\boldsymbol{f}_c$ for target batch c. Its entries are the number of loops, external momenta, inverse propagators, physical propagators and symbolic invariants; $\log(1+n_c)$, where n_c is the number of targets; the maximum dot count, maximum numerator rank, mean dot count and mean numerator rank; the number of target sectors; and seven indicators for the graph structures. An indicator is one for its graph and zero for the other graphs. There are eighteen entries in total.

For an index vector $\boldsymbol{a}$, define

$$r(\boldsymbol{a}) = \sum_i \max(a_i, 0), \qquad s(\boldsymbol{a}) = \sum_i \max(-a_i, 0), \qquad q(\boldsymbol{a}) = \sum_i \max(a_i - 1, 0). \tag{58}$$

Here r is the positive index sum, s is the numerator rank, and q is the number of extra denominator powers, called dots. These definitions follow directly by counting the denominator and numerator factors. The initial bounds $B_c = (r_c, s_c, q_c)$ are the componentwise maxima over the targets. The trial seed regions use B_c, $B_c - (1,1,1)$, $B_c - (1,0,1)$, or $B_c + (1,1,1)$. Negative bounds are set to zero. These choices restrict equation generation. They do not assert that a smaller region always supplies a complete reduction.

Let t_{cj} be the median complete time of strategy j on training batch c, and let t_{c0} be the time of the integrated native strategy. The training response is

$$y_{cj} = \log \frac{t_{cj}}{t_{c0}}. \tag{59}$$

Each time includes unsuccessful attempts and the successful fallback. Thus the response penalizes a strategy that omits required equations. A smaller response means a smaller ratio because the logarithm is strictly increasing on positive numbers.

We fit constant, tree, linear and nearest-neighbour predictors of this response. The constant predictor averages each strategy's training responses. A tree splits the feature space and uses a constant vector within each leaf, following regression-tree construction [12]. For a leaf containing training set A, minimizing $\sum_{c\in A}(y_{cj} - v_j)^2$ gives

$$\frac{\partial}{\partial v_j} \sum_{c\in A} (y_{cj} - v_j)^2 = 2|A| v_j - 2 \sum_{c\in A} y_{cj} = 0, \qquad v_j = \frac{1}{|A|} \sum_{c\in A} y_{cj}. \tag{60}$$

Each split minimizes the sum of squared errors over the seven strategies. The tested maximum depths are two, four and six. Each depth is tested with a minimum of either one or three training batches per child.

For the linear models, we subtract each feature's training mean and divide by its training population standard deviation, computed with divisor equal to the number of training batches. A constant feature uses divisor one. Let Z contain these standardized feature rows with an leading intercept column, and let Y contain the responses. Ridge regression minimizes

$$J(W) = \|ZW - Y\|_F^2 + \lambda \operatorname{tr}(W^T P W), \qquad P = \operatorname{diag}(0, 1, \ldots, 1). \tag{61}$$

Here W is the matrix of fitted coefficients, $\|\cdot\|_F^2$ is the sum of squared matrix entries, and P leaves the intercept unpenalized. Differentiating each entry of W gives

$$\nabla_W J = 2Z^T(ZW - Y) + 2\lambda P W = 0, \qquad (Z^T Z + \lambda P) W = Z^T Y. \tag{62}$$

We solve this system for $\lambda = 0.1, 1, 10$.

The nearest-neighbour predictors restrict neighbours to the same graph, loop count, external-momentum count, propagator counts and number of symbolic invariants. They choose up to

two or four training batches with the smallest squared distance in standardized features. The predicted response is the mean of their response vectors. The leaf-mean derivative above also derives this local constant estimate. If no eligible neighbour exists, the predictor returns equal costs and the selector uses native choices.

Let j_* be the strategy with the smallest predicted logarithmic cost $\widehat{y}_j(\boldsymbol{f})$. Validation selects a margin $\gamma \in \{0, 0.05, 0.15\}$, and the model chooses the strategy as follows:

$$\widehat{j}(\boldsymbol{f}) = \begin{cases} j_*, & \widehat{y}_{j_*}(\boldsymbol{f}) < \widehat{y}_0(\boldsymbol{f}) - \gamma, \\ 0, & \text{otherwise.} \end{cases} \tag{63}$$

Subtracting the two predicted logarithmic costs shows that the condition requires a predicted relative cost below $e^{-\gamma}$. Validation selects the model and margin with the smallest geometric mean time ratio. The selected settings remain fixed during reserved tests. This finite comparison establishes the best measured candidate within the study. It does not prove a globally optimal reduction strategy.

I.1 Parameter points and target lists

The complete-cost study keeps the seven denominator routings and their kinematic conventions defined earlier. Let h identify a rational parameter profile. Training uses $h = 41, 42$, validation uses $h = 43$, and testing uses $h = 44, 45$. In a common squared-momentum unit, the sunrise has squared masses $(2, 7, 11)$ and $p^2 = -1009 - 12h$. The planar box has $(s, t) = (-1009 - 12h, -1021 - 14h)$. The three-loop vacuum has squared masses $(1, 1, 1, 1, 1, 53 + 2h)$. The ladder keeps squared masses $(1, 0, 0, 0, 0, 0, 0, 0)$ and has $p^2 = -1009 - 12h$. The scattering assignments are

$$(s, t)_{\mathrm{NP}} = (-113 - 18h, -127 - 10h), \tag{64}$$

$$(s_{12}, s_{23}, s_{34}, s_{45}, s_{51})_{\mathrm{P}} = (-11 - 20h, -13 - 20h, -17 - 20h, -19 - 20h, -23 - 20h), \tag{65}$$

$$(s, t, v)_{\mathrm{box}} = (-113 - 18h, -127 - 10h, -149 - 12h). \tag{66}$$

These equations specify the selected input values. The dimension remains symbolic. The symbolic batches keep all independent invariants of their scattering graph symbolic.

For the four earlier multiloop graphs, we retain the ordered target list that starts with the corner, raises each physical denominator power once, and then introduces each single auxiliary numerator. We repeat these steps after removing one, two and three physical denominators, retain at least two, and remove duplicates. Each rational profile supplies a four-target batch and a batch of the first 32 targets, or all 21 targets for the sunrise. The three scattering graphs use the same construction with at most one removed denominator. Their larger batches have 32, 31 and 21 targets for the nonplanar box, pentagon and virtuality box.

For the symbolic nonplanar box and pentagon, training takes target entries 12–15, validation takes 16–19, and testing takes 20–23, with numbering starting at zero. These lists do not overlap. For the symbolic virtuality box, the respective single targets are $(3, 1, 1, 1)$, $(1, 3, 1, 1)$ and $(1, 1, 3, 1)$. These targets have two extra powers on one physical denominator. The rational batches reuse target patterns at different parameter points.

The joint-request test uses the larger rational batch at $h = 44$ for each graph and the three symbolic test batches. The symbolic virtuality box also includes $(1, 1, 1, 3)$, so it can supply two disjoint requests. A group of n targets is divided after entry $\lfloor n/2 \rfloor$, with the first request containing that many entries. The two requests have identical family definitions and are available together. Their union contains each target once. Linearity of the reduction lets us return the coefficients for each original request by selecting its target rows from the joint result. This changes no coefficient and requires no additional integral identity. The test reuses reserved parameter points and keeps the learned model fixed.

I.2 Combining the measured times

Let m_{cj} be the median of the five test times for batch c and method j, and let method zero be its stated native control. For graph g with batch set C_g, we define

$$R_{gj} = \exp\left[\frac{1}{|C_g|}\sum_{c\in C_g}\log\frac{m_{cj}}{m_{c0}}\right], \qquad R_j = \exp\left[\frac{1}{7}\sum_{g=1}^{7}\log R_{gj}\right]. \tag{67}$$

The logarithm of a product equals the sum of the logarithms, so each expression is the corresponding geometric mean of the time ratios. The second expression assigns equal weight to each graph even when the numbers of batches differ. A ratio $R_j < 1$ means less time. The percentage saving is $100(1 - R_j)$, and the reciprocal $1/R_j$ is the speedup.

For the descriptive intervals, we sample seven graph labels with replacement, calculate their mean logarithmic ratio, and exponentiate. We repeat this 10,000 times and report the 2.5th and 97.5th percentiles. This procedure describes variation among the seven tested graphs. It does not provide a population guarantee for other integral families. The joint-request comparison uses the same calculation, with separate native requests as its first control. Its parameter points overlap the main reserved test, so it is an additional comparison on the same test population.

J Separate cost models for each graph

This experiment retains the seven denominator definitions and profile formulas in Appendix I. Development uses profile $h = 61$, training uses $h = 62, 63$, validation uses $h = 64$, and the reserved test uses $h = 65, 66$. The dimension remains symbolic. Related target batches share the same split, so their rational kinematic values cannot pass from the training set into the reserved set. The graph structures and target patterns occur in both sets.

Let P be the number of physical propagators and let $\boldsymbol{b} = (1, \ldots, 1, 0, \ldots, 0)$ have P entries equal to one. Let $\boldsymbol{e}_j$ be the unit vector for index j. The first batch contains $\boldsymbol{b} + 2\boldsymbol{e}_j$ for $j = 1, \ldots, \min(P, 3)$. Thus one denominator has power three in each target. For the second batch, define the lower-sector corner

$$\boldsymbol{b}^{\mathrm{low}} = \begin{cases} \boldsymbol{b} - \boldsymbol{e}_1, & P < 5, \\ \boldsymbol{b} - \boldsymbol{e}_1 - \boldsymbol{e}_3, & P \geq 5. \end{cases} \tag{68}$$

Include this corner and up to three single-dot targets, $\boldsymbol{b}^{\mathrm{low}} + \boldsymbol{e}_j$ with $b_j^{\mathrm{low}} = 1$, taking the first available indices in increasing order. The third batch uses the original corner $\boldsymbol{b}$ and contains $\boldsymbol{b} - 2\boldsymbol{e}_j$ for the first three auxiliary entries, or all auxiliary entries when fewer than three exist. When a family has no auxiliary entry, this construction repeats the first batch and we remove the duplicate. These formulas define the inputs; they do not infer their coefficients from the masters.

J.1 The allowed changes to Kira

For target batch $\mathcal{T}$ and a target $\boldsymbol{a}$, let $S(\boldsymbol{a}) = \{i : a_i > 0\}$ be its positive-index set. One strategy supplies only maximal target sectors:

$$\mathcal{S}_{\max} = \{S(\boldsymbol{a}) : \boldsymbol{a} \in \mathcal{T},\ S(\boldsymbol{a}) \not\subseteq S(\boldsymbol{a}') \text{ for every } \boldsymbol{a}' \in \mathcal{T}\}. \tag{69}$$

Kira then includes their subsectors. To see why this is a useful candidate, consider the shifted integrals produced by the IBP derivative. A raised index $a_i + 1$ carries a factor a_i, and the other shifts lower an index. If $a_i = 0$, the raising term vanishes; if $a_i < 0$, raising it once leaves it nonpositive. Thus an IBP equation seeded in one sector stays in that sector or its subsectors.

This observation does not prove that the restricted pool contains every relation useful for a chosen master basis. We therefore retain the basis check and full-sector fallback.

Another strategy uses Kira's numerator truncation [5]. In a sector with t positive indices, Kira restricts the inherited numerator bound by

$$s \leq \max(1, t - \tau + 1, s_{\text{explicit}}), \tag{70}$$

where τ is the chosen integer cutoff and s_{explicit} is a bound explicitly supplied for that sector. We test $\tau = P - s_c$ and $\tau = P - s_c + 1$, where s_c is the largest target numerator rank. The top-level sector retains its explicit target bound. The first cutoff combined with maximal target sectors forms the deterministic control. The development probes also test $\tau = P - s_c - 1$.

The remaining choices use the original bounds B_c, the smaller bounds $B_c - (1, 0, 1)$ with negative entries replaced by zero, the larger bounds $B_c + (1, 1, 1)$, Kira integral ordering five [13], or the preceding trained equation filter. Together these choices give nine strategies. Before each attempt, every strategy enforces a nonempty seed region in its declared top-level sectors. If $\mathcal{S}_0$ is that sector list and r_{req} is the requested positive-index bound, the actual bound is

$$r_{\text{run}} = \max\left(r_{\text{req}}, \max_{S \in \mathcal{S}_0} |S|\right). \tag{71}$$

Each positive index is at least one, so a seed in sector S has $\sum_{i \in S} a_i \geq |S|$. This proves the lower bound used in Eq. (71). The correction prevents an invalid configuration from adding a failed startup to any method's cost. Identical input configurations share one training or validation measurement. If a proposed reduction fails, it retries with native choices at B_c and then at $B_c + (1, 1, 1)$. The recorded cost includes every attempt.

J.2 Fitting and selecting the models

Each graph has its own predictors of $y_{cj} = \log(t_{cj}/t_{c0})$, where t_{cj} is the median complete cost of strategy j on batch c and zero denotes native choices. The nine input features are the target count; maximum dot count and numerator rank; minimum and maximum positive-index counts; mean dot count and numerator rank; number of maximal target sectors; and number of physical propagators. These features use the requested indices and family definition. They contain no reduced coefficients.

We fit ten candidates per graph: one constant predictor, trees of maximum depth one, two and three, a bagged ensemble of sixteen depth-three trees, ridge regressions with penalties $0.1, 1, 10$, and nearest-neighbour predictors with one and three neighbours. Trees use squared-error splits with at least one row in each child. Each ensemble tree receives a bootstrap sample of the training batches and uses all features at each split. This follows bootstrap aggregation [16]. For fitted trees $f_1, \ldots, f_{16}$, the ensemble prediction is

$$\widehat{y}_j(\boldsymbol{f}) = \frac{1}{16} \sum_{b=1}^{16} f_{bj}(\boldsymbol{f}). \tag{72}$$

This average also follows by minimizing $\sum_b (f_{bj}(\boldsymbol{f}) - v_j)^2$ with respect to v_j, using the leaf-mean derivative in Appendix I. That appendix derives the ridge fit and the local-mean estimates used here. Neighbour distances use the training standard deviations; a constant feature uses divisor one. We use two timing repetitions for training and three for validation. Within each batch comparison, every strategy uses the same computer. Different training batches can use different computers, and the training response normalizes each cost to its own batch baseline.

Validation considers decision margins $0, 0.05, 0.15$ in Eq. (63). It chooses the candidate and margin with the lowest complete cost separately for each graph. The controls are native choices, the deterministic sector-and-truncation rule, and the single strategy with the lowest geometric

mean training cost for that graph. A fixed strategy can include the older equation filter. We enable the new model only when its validation cost ratio to the fastest control is at most 0.95. Otherwise, the route uses that control. This threshold is a selection rule, not a significance test. Unknown graph structures and features outside the fitted range use native choices.

The reserved comparison uses five repetitions of native choices, the deterministic rule, the fixed strategy per graph, and the selected route. The order rotates across batches and repetitions, giving every method the same number of appearances in each run position over the full test. Each run uses the same targets, prescribed basis, executable, and one processor. All methods retain the preceding exact implementation improvements, so the native control uses that same implementation with native selection rules. Full time includes input preparation, prediction, native IBP generation, exact reduction, coefficient checks, cleanup, and retries. Offline fitting, reference discovery, and saving measurement records stay outside the timer. Separate application checks receive the basis without target coefficients and compare their answers only after completion. The calling program and fitted parameters are loaded before the timing trials, and each trial starts a fresh Kira process. The additional application checks also record time with a newly started calling program; they do not supply a matched speed comparison for that condition.

We combine five-run median ratios within each graph and then give the seven graphs equal weight, as derived in Appendix I. The aggregate intervals resample graph groups. Per-graph plot bars show the range across the two reserved kinematic profiles. Two profiles do not support a general significance claim for that integral type, so those bars are reported as ranges.

References

[1] Stefan Weinzierl. *Feynman Integrals: A Comprehensive Treatment for Students and Researchers*. UNITEXT for Physics. Springer, 2022. arXiv:2201.03593; publisher DOI.

[2] K. G. Chetyrkin and F. V. Tkachov. Integration by parts: The algorithm to calculate β-functions in 4 loops. *Nuclear Physics B*, 192:159–204, 1981. publisher DOI.

[3] Stefano Laporta. High-precision calculation of multiloop Feynman integrals by difference equations. *International Journal of Modern Physics A*, 15:5087–5159, 2000. arXiv:hep-ph/0102033.

[4] Philipp Maierhöfer, Johann Usovitsch, and Peter Uwer. Kira—a Feynman integral reduction program. *Computer Physics Communications*, 230:99–112, 2018. arXiv:1705.05610.

[5] Fabian Lange, Johann Usovitsch, and Zihao Wu. Kira 3: Integral reduction with efficient seeding and optimized equation selection. *Computer Physics Communications*, 322:109999, 2026. arXiv:2505.20197.

[6] Zhuo-Yang Song, Tong-Zhi Yang, Qing-Hong Cao, Ming-xing Luo, and Hua Xing Zhu. Explainable AI-assisted optimization for Feynman integral reduction, 2025. arXiv:2502.09544.

[7] Jonas Klappert, Fabian Lange, Philipp Maierhöfer, and Johann Usovitsch. Integral reduction with Kira 2.0 and finite field methods. *Computer Physics Communications*, 266:108024, 2021. arXiv:2008.06494.

[8] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov, and Alexander Smola. Deep sets. In *Advances in Neural Information Processing Systems*, volume 30, 2017. arXiv:1703.06114.

[9] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017. arXiv:1706.03762.

[10] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. arXiv:1711.05101.

[11] Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432, 2015.

[12] Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. *Classification and Regression Trees*. Chapman & Hall, 1984. Publisher record.

[13] Philipp Maierhöfer and Johann Usovitsch. Kira 1.2 release notes, 2018. arXiv:1812.01491.

[14] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. publisher DOI.

[15] Richard Bellman. On the theory of dynamic programming. *Proceedings of the National Academy of Sciences*, 38(8):716–719, 1952. publisher DOI.

[16] Leo Breiman. Bagging predictors. *Machine Learning*, 24:123–140, 1996. Publisher record.